

Security Threats and Mitigation Strategies for Hybrid Active
Directory Architectures: A Qualitative Case Study Analysis of
Cross-Architecture Attack Propagation.

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Abstract

Hybrid Active Directory (AD) architectures, integrating on-premises Windows Server AD with Microsoft Entra ID via Azure AD Connect are now used by 78% of enterprises, yet remain critically vulnerable to cross-architecture attack propagation. This qualitative multiple-case study analyzes 19 publicly documented breaches (2017–2025) to investigate how attackers exploit synchronization mechanisms, abuse credentials across trust boundaries, and circumvent defensive controls. Findings reveal that Azure AD Connect servers, synchronization rules, and object writeback features serve as primary attack conduits, enabling bidirectional escalation between cloud and on-premises domains. Credential abuse patterns, including Pass-through Authentication (PTA) agent manipulation (“Double Agent” vulnerability), NTLM hash reuse, and SAML token forgery demonstrate that identities are treated as universal tokens across architectures.

Mitigation strategies such as Multi-Factor Authentication (MFA) and Conditional Access Policies (CAPs) frequently failed due to inconsistent enforcement, while Privileged Access Management (PAM) and network segmentation showed greater efficacy when implemented holistically. The study proposes a hybrid-specific attack propagation taxonomy and offers actionable recommendations grounded in Zero Trust principles. By synthesizing real-world incident data, this research addresses a critical gap in the literature and provides enterprise security teams with evidence-based guidance to harden hybrid identity infrastructures against modern, cross-boundary threats.

Table of contents

CHAPTER 1: INTRODUCTION.	12
1.1 BACKGROUND OF THE STUDY.....	12
1.2 RESEARCH PROBLEM.....	13
1.3 RESEARCH OBJECTIVES AND QUESTIONS.....	14
1.4 SIGNIFICANCE OF THE STUDY.....	14
1.5 SCOPE AND LIMITATIONS.	15
1.5.1 Scope.....	15
1.5.2 Limitations.	15
CHAPTER 2: LITERATURE REVIEW.	16
2.1 INTRODUCTION.....	16
2.2 AZURE AD CONNECT ARCHITECTURE AND THE HYBRID TRUST MODEL.....	16
2.3 SYNCHRONIZATION RISKS: FROM HASH SYNC TO OBJECT WRITEBACK.....	18
2.4 CROSS-BOUNDARY ATTACK PROPAGATION: GOLDEN TICKETS TO CLOUD ADMIN.....	19
2.5 SOCIAL ENGINEERING IN HYBRID CONTEXTS: THE HUMAN BRIDGE.....	20
2.6 MITIGATION MODELS: ZERO TRUST AS THE UNIFYING FRAMEWORK.....	20
2.7 IDENTIFICATION OF GAPS IN THE LITERATURE	22
2.8 CONCEPTUAL FRAMEWORK.....	22
CHAPTER 3: METHODOLOGY	23
3.1 INTRODUCTION.....	23
3.2 RESEARCH DESIGN AND PHILOSOPHICAL POSITIONING.....	23
3.2.1 Multiple-Case Study Design	23
3.2.2 Epistemological Positioning.....	24
3.3 Case Selection and Sampling Strategy.....	25
3.3.1 Inclusion Criteria	25
3.3.2 Sample Composition	25
3.4 DATA COLLECTION FROM SECONDARY SOURCES	26
3.4.1 Source Types and Quality Assessment	26
3.4.2 Data Extraction Protocol.....	27
3.5 CONCEPTUAL FRAMEWORK-DRIVEN THEMATIC ANALYSIS.....	28
3.5.1 The Conceptual Framework.....	28
3.5.2 Five-Phase Analytical Process.....	29
3.6 QUALITY ASSURANCE AND METHODOLOGICAL RIGOR.....	33
3.6.1 CREDIBILITY (INTERNAL VALIDITY).....	33
3.6.2 Transferability (External Validity)	34
3.6.3 Dependability (Reliability).....	34
3.6.4 Confirmability (Objectivity).....	34
3.7 ETHICAL CONSIDERATIONS	35
3.8 METHODOLOGICAL LIMITATIONS AND CONSTRAINTS.....	36
CHAPTER 4: FINDINGS	37
4.1 INTRODUCTION.....	37
4.2 RQ1: EXPLOITATION OF SYNCHRONIZATION MECHANISMS IN CROSS-ARCHITECTURE ATTACK PROPAGATION.....	39
4.2.1 Azure AD Connect Servers as Privileged Attack Conduits	40
4.2.2 Synchronization Rule Manipulation and Malicious Object Injection	41
4.2.3 Object Writeback as Bidirectional Escalation Vector	42
4.2.4 Synthesis: Synchronization as Primary Attack Vector.....	43
4.3 RQ2: CROSS-ARCHITECTURE CREDENTIAL ABUSE VULNERABILITIES.....	44

4.3.1 Pass-Through Authentication Agent Exploitation.....	45
4.3.2 NTLM Hash Reuse Across Architectural Boundaries.....	46
4.3.3 Token-Based Attacks via Federated Trust Exploitation.....	47
4.3.4 Synthesis: Credential Universality as Systemic Vulnerability	48
4.4 RQ3: MITIGATION STRATEGY EFFICACY IN HYBRID ENVIRONMENTS.....	49
4.4.1 MFA and Conditional Access Policy Failures.....	49
4.4.2 High-Efficacy Controls: PAM, Segmentation, and Monitoring	52
4.4.3 Synthesis: The Efficacy-Deployment Gap	53
4.5 CROSS-CASE SYNTHESIS: A TAXONOMY OF HYBRID AD ATTACK PROPAGATION	54
CHAPTER 5: DISCUSSION	57
5.1 INTRODUCTION.....	57
5.2 INTERPRETATION OF RESULTS	57
5.2.1 Synchronization as the Dominant Attack Vector (RQ1).....	57
5.2.2 CREDENTIAL UNIVERSALITY AS ARCHITECTURAL VULNERABILITY (RQ2).....	59
5.2.3 The Efficacy-Deployment Gap and Control Selection Failures (RQ3)	60
5.2.4 IDENTITY THREAT DETECTION AND RESPONSE (ITDR) AS SYSTEMIC MITIGATION.....	62
5.3 COMPARISON WITH EXISTING LITERATURE	63
5.3.1 Validation of Hybrid-Specific Vulnerability Research.....	63
5.3.2 Divergence from Traditional AD Security Models	63
5.3.3 Social Engineering in Hybrid Contexts.....	64
5.3.4 Zero Trust Implementation Gaps.....	65
5.4 THEORETICAL AND PRACTICAL IMPLICATIONS.....	66
5.4.1 Theoretical Contributions	66
5.4.2 Practical Implications for Security Practitioners	66
5.5 LIMITATIONS OF THE STUDY.....	68
5.5.1 Methodological Limitations.....	68
5.5.2 Potential Biases and Confounding Factors.....	69
5.5.3 Generalizability Constraints.....	69
5.6 RECOMMENDATIONS FOR FUTURE RESEARCH.....	70
5.6.1 Priority Research Directions	70
5.6.2 Methodological Extensions	71
5.6.3 Emerging Technology Implications.....	72
5.7 CHAPTER SUMMARY	72
CHAPTER 6: CONCLUSION	74
6.1 SUMMARY OF KEY FINDINGS.....	74
6.2 CONTRIBUTIONS TO KNOWLEDGE.....	75
6.3 PRACTICAL IMPLICATIONS.....	76
6.4 FINAL THOUGHTS	77

Table of figures

Figure 1: reference architecture shows best practices for integrating on-premises Active Directory domains with Microsoft Entra ID	17
Figure 2: PTA and Identity Synchronization Architecture	18
Figure 3: Core Zero Trust Logical Components, NIST	21
Figure 4: Research Workflow and Analytical Process	32

Table of tables

Table no.	Description	Page no.
Table 1	Overview of Analyzed Hybrid AD Breach Cases (2017-2025)	39
Table 2	Azure AD Connect Server Exploitation Patterns	41
Table 3	Synchronization Exploitation Taxonomy by Attack Direction	43
Table 4	Credential Abuse Patterns and Reuse Vectors	47
Table 5	Mitigation Strategy Deployment and Effectiveness Outcomes	51
Table 6	Five-Phase Hybrid AD Attack Progression Model	56

List of abbreviations

- **AD** - Active Directory
- **ADFS** - Active Directory Federation Services
- **API** - Application Programming Interface
- **APT** - Advanced Persistent Threat
- **CAP** - Conditional Access Policy
- **CISA** - Cybersecurity and Infrastructure Security Agency
- **DC** - Domain Controller
- **DRS** - Directory Replication Service
- **EDR** - Endpoint Detection and Response
- **GDPR** - General Data Protection Regulation
- **HIPAA** - Health Insurance Portability and Accountability Act
- **IAM** - Identity and Access Management
- **ITDR** - Identity Threat Detection and Response
- **MFA** - Multi-Factor Authentication
- **MITRE ATT&CK** - MITRE Adversarial Tactics, Techniques, and Common Knowledge
- **NIST** - National Institute of Standards and Technology

- **NTLM** - New Technology LAN Manager
- **PAM** - Privileged Access Management
- **PHS** - Password Hash Synchronization
- **PTA** - Pass-Through Authentication
- **SAML** - Security Assertion Markup Language
- **SEC** - Securities and Exchange Commission
- **SIEM** - Security Information and Event Management
- **TGT** - Ticket Granting Ticket
- **ZTA** - Zero Trust Architecture

Glossary

- **Active Directory (AD):** Microsoft's directory service for Windows domain networks, providing centralized authentication, authorization, and resource management. AD uses hierarchical structures of domains, organizational units, and objects to manage identities and access control across enterprise environments.
- **Azure AD Connect:** Microsoft's synchronization tool that integrates on-premises Active Directory with cloud-based Microsoft Entra ID (formerly Azure AD). It establishes a trust bridge between environments through Password Hash Synchronization, Pass-through Authentication, or Active Directory Federation Services.

- **Bidirectional Escalation:** Attack technique where adversaries leverage synchronized identities and trust relationships to move laterally between on-premises and cloud environments in both directions, exploiting the mutual trust established by hybrid architectures.
- **Conditional Access Policy (CAP):** Cloud-based security controls in Microsoft Entra ID that enforce access requirements based on conditions such as user location, device compliance, application sensitivity, and risk level. CAPs enable granular control over authentication and authorization decisions.
- **Cross-Architecture Attack Propagation:** The process by which attackers exploit synchronization mechanisms, shared credentials, and trust relationships to move between on-premises Active Directory and cloud-based Microsoft Entra ID environments, effectively traversing architectural boundaries.
- **Credential Abuse:** Unauthorized use of legitimate credentials, tokens, or authentication materials to gain access to systems or data. In hybrid environments, this includes reusing NTLM hashes, manipulating Pass-through Authentication agents, and forging SAML tokens.
- **DCSync Attack:** Attack technique that abuses Directory Replication Service (DRS) permissions to remotely extract password hashes and sensitive data from domain controllers. Attackers simulate domain controller replication requests to harvest credentials without direct access to DC servers.
- **Double Agent Vulnerability:** Security flaw in Pass-through Authentication configurations where attackers with control over PTA agent servers can manipulate authentication flows to bypass credential verification, enabling authentication as any synchronized user without valid credentials.

- **Golden Ticket Attack:** Advanced persistence technique involving the forgery of Kerberos Ticket Granting Tickets (TGTs) using the compromised KRBTGT account hash. In hybrid environments, these forged tickets can grant persistent access to both on-premises resources and synchronized cloud accounts.
- **Hybrid Active Directory:** Enterprise identity architecture integrating on-premises Windows Server Active Directory with cloud-based Microsoft Entra ID through synchronization mechanisms. This architecture enables unified identity management across on-premises and cloud resources while creating potential security vulnerabilities at trust boundaries.
- **Identity Threat Detection and Response (ITDR):** Specialized security discipline focused on detecting, investigating, and responding to identity-based attacks. ITDR solutions monitor authentication patterns, privileged access, and synchronization activities to identify anomalous behavior indicative of credential abuse or account compromise.
- **Kerberoasting:** Attack technique targeting service accounts in Active Directory by requesting Kerberos service tickets encrypted with service account password hashes. Attackers capture these tickets and perform offline brute-force attacks to recover plaintext passwords.
- **Lateral Movement:** Post-compromise technique where attackers move horizontally through a network or across environments using compromised credentials or exploited trust relationships to access additional systems, escalate privileges, and expand their foothold.
- **Microsoft Entra ID:** Microsoft's cloud-based identity and access management service (formerly Azure Active Directory). Entra ID provides authentication and authorization for cloud applications, supports hybrid identity scenarios through Azure AD Connect, and offers advanced security features including conditional access and identity protection.

- **Multi-Factor Authentication (MFA):** Security control requiring users to provide two or more verification factors (something they know, have, or are) to authenticate. Common implementations include passwords combined with SMS codes, authenticator apps, biometrics, or hardware tokens.
- **Object Writeback:** Azure AD Connect feature enabling cloud-created objects (groups, devices, users) to be synchronized back to on-premises Active Directory. When exploited, attackers can create privileged cloud groups that inherit on-premises permissions, enabling cloud-to-on-premises privilege escalation.
- **Password Hash Synchronization (PHS):** Hybrid authentication method where NTLM password hashes are synchronized from on-premises Active Directory to Microsoft Entra ID. PHS enables cloud authentication while maintaining a copy of password hashes in both environments, creating potential credential reuse vectors if compromised.
- **Pass-Through Authentication (PTA):** Hybrid authentication method using lightweight agents installed on-premises to relay authentication requests from cloud services to on-premises domain controllers in real-time. Unlike PHS, password hashes remain exclusively on-premises, but PTA agents become high-value attack targets.
- **Privileged Access Management (PAM):** Security framework and toolset for controlling, monitoring, and auditing privileged accounts and access. PAM solutions enforce just-in-time access, time-bound privileges, approval workflows, and session monitoring to reduce standing administrative access and credential exposure.
- **Privilege Escalation:** Attack progression phase where adversaries exploit vulnerabilities, misconfigurations, or compromised credentials to obtain higher-level permissions than initially granted. In hybrid environments, this often involves targeting synchronization infrastructure to gain administrative access across both domains.

- **SAML Token:** Security Assertion Markup Language authentication token used in federated identity scenarios. SAML tokens contain assertions about user identity and attributes, signed by an identity provider. When forged using compromised signing certificates, these tokens enable persistent unauthorized access.
- **Synchronization Abuse:** Exploitation of identity synchronization mechanisms to propagate attacks between on-premises and cloud environments. Techniques include compromising Azure AD Connect servers, manipulating sync rules, injecting malicious objects, and exploiting writeback features.
- **Tier 0 Asset:** Critical infrastructure components with direct or effective administrative control over identity systems. Examples include domain controllers, Azure AD Connect servers, certificate authorities, and PAM systems. Compromise of Tier 0 assets enables complete environment takeover.
- **Zero Trust Architecture (ZTA):** Security framework based on the principle of 'never trust, always verify.' ZTA assumes no implicit trust based on network location, requires continuous verification of all access requests, enforces least-privilege access, and implements micro-segmentation. Critical for securing hybrid identity environments where trust boundaries are blurred.

Chapter 1: Introduction.

1.1 Background of the Study.

Microsoft Active Directory (AD), which remains the de facto standard for identity and access management across global organizations. Approximately 90% of Fortune 1000 companies rely on AD to authenticate users, authorize access, and manage digital identities (CrowdStrike, 2025). As digital transformation accelerates, enterprises increasingly adopt **hybrid AD architectures** integrating on-premises Windows Server AD with Microsoft Entra ID (formerly Azure AD) to balance legacy system compatibility with cloud agility.

This architectural convergence is now the norm rather than the exception. By 2024, 78% of enterprises operated hybrid AD environments, leveraging synchronization tools such as **Azure AD Connect** to bridge on-premises and cloud identity domains (Diericx, 2024; Semperis, 2023; Microsoft, 2024). While this model offers operational flexibility, it also introduces complex interdependencies that expand the attack surface and create novel pathways for threat propagation. Unlike isolated on-premises or cloud-only deployments, hybrid environments inherit vulnerabilities from both domains while introducing unique risks at their integration points, particularly in credential synchronization, authentication relay mechanisms, and policy enforcement boundaries.

Recent high-impact breaches underscore the severity of these risks. The 2023 Azure AD conditional access policy tampering incident exposed 18 million enterprise accounts to persistent backdoor access (Secureworks, 2023). In 2024, Russian state-sponsored actors breached Microsoft's own corporate email systems by exploiting a legacy test tenant and pivoting through hybrid identity pathways (Microsoft Security Response Center, 2024). Similarly, the MGM Resorts and Caesars Entertainment cyberattacks in 2023 demonstrated how threat actors like **Scattered Spider** leverage social engineering to

compromise cloud identity portals and subsequently gain control over on-premises infrastructure via legitimate synchronization channels. (Mandiant, 2024; Krebs, 2023)

These incidents reveal a critical insight: **attackers no longer target on-premises or cloud AD in isolation**. Instead, they exploit the **interstitial space** between architectures where trust is assumed, visibility is fragmented, and defensive controls are inconsistently applied.

1.2 Research Problem.

Despite the widespread adoption of hybrid AD, academic and industry research remains siloed. Existing studies predominantly analyze on-premises AD vulnerabilities (Kumar et al., 2024; Syynimaa, 2020) (e.g., Kerberoasting, Golden Ticket attacks) or cloud-specific risks (e.g., API abuse, OAuth misconfigurations) in isolation (Kumar et al., 2024; Syynimaa, 2020). Even when hybrid environments are acknowledged, the focus is often on component-level hardening (e.g., securing Azure AD Connect servers) rather than systemic attack propagation dynamics.

Consequently, a significant knowledge gap persists: how do adversaries move laterally across architectural boundaries in hybrid AD, and what enables this cross-platform escalation? Current security frameworks lack a unified understanding of:

- The role of synchronization mechanisms as attack conduits,
- The reuse and abuse of credentials across trust domains,
- The failure modes of mitigation strategies in real-world hybrid compromises.

This gap leaves security practitioners without evidence-based models to anticipate, detect, or contain cross-architecture threats. (Chen et al., 2024)

1.3 Research Objectives and Questions.

- RQ1: How do attackers exploit synchronization mechanisms in hybrid AD environments to propagate attacks across architectural boundaries?
- RQ2: What common vulnerabilities enable cross-architecture credential abuse in hybrid AD implementations?
- RQ3: Which mitigation strategies effectively address hybrid-specific attack vectors based on documented incidents?

These questions focus on **attack propagation logic**, **vulnerability patterns**, and **defensive efficacy** all derived from empirical breach data.

1.4 Significance of the Study.

This dissertation contributes to both academic knowledge and practical cybersecurity practice. Theoretically, it advances the field of identity security by proposing a cross-architecture threat model grounded in real-world incident data. It responds to calls in the literature for integrated analyses that transcend traditional on-premises/cloud dichotomies (Chen et al., 2024; CounterCraft Security, 2025).

Practically, the findings offer actionable insights for enterprise security teams managing hybrid AD. By identifying recurring attack patterns, such as Azure AD Connect server compromise, Pass-through Authentication (PTA) agent manipulation, and federated identity hijacking, the study enables organizations to prioritize controls that disrupt actual adversary behaviors. Furthermore, the evaluation of mitigation failures (e.g., MFA bypasses, ineffective conditional access policies) provides evidence-based guidance for improving defense-in-depth strategies.

Given that hybrid AD is now the dominant enterprise architecture, this research addresses a **critical and timely security challenge** with direct relevance to global organizations.

1.5 Scope and Limitations.

1.5.1 Scope.

This study focuses exclusively on **hybrid Active Directory environments** that integrate Windows Server AD (2016 or later) with Microsoft Entra ID via official synchronization mechanisms, including:

- Password Hash Synchronization (PHS),
- Pass-through Authentication (PTA),
- Active Directory Federation Services (ADFS).

The analysis centers on **attack propagation pathways** that traverse the on-premises/cloud boundary, with particular attention to synchronization infrastructure, credential flows, and privilege escalation chains.

1.5.2 Limitations.

1. **Secondary Data Only:** The research relies entirely on publicly reported breach data, vendor advisories, and academic literature. No primary data collection (e.g., interviews, surveys, or log analysis from live environments) was conducted.
2. **No Experimental Validation:** Due to resource and methodological constraints, findings are not validated through lab simulations or tool-based detection.
3. **Reporting Bias:** Publicly disclosed incidents may overrepresent high-profile breaches and underrepresent stealthy or unreported compromises.
4. **Exclusion of Non-Microsoft Ecosystems:** Third-party identity providers, non-Windows endpoints, and alternative directory services are outside the scope.

These limitations are acknowledged transparently and do not undermine the validity of a qualitative case study design, which prioritizes depth and contextual understanding over generalizability (Yin, 2018).

Chapter 2: Literature Review.

2.1 Introduction.

This chapter critically reviews existing literature on security threats and mitigation strategies in hybrid Active Directory (AD) environments. The focus is deliberately narrowed to **hybrid architectures** defined as integrations of on-premises Windows Server AD with Microsoft Entra ID (formerly Azure AD) via synchronization mechanisms such as Azure AD Connect reflecting the scope refinement advised during the mid-point review. While prior research has examined on-premises and cloud AD in isolation, this review synthesizes findings that specifically address the **interstitial attack surface** created by architectural convergence. The goal is to establish a conceptual foundation for analyzing cross-architecture attack propagation, identify persistent knowledge gaps, and justify the need for a unified, evidence-based threat model grounded in real-world breach data.

2.2 Azure AD Connect Architecture and the Hybrid Trust Model.

Hybrid AD environments rely on **Azure AD Connect** as the central synchronization engine, establishing a unidirectional or bidirectional trust bridge between on-premises and cloud identity domains. As Diericx (2024) emphasizes, Azure AD Connect is not merely a data replicator; it is a **privileged identity conduit** that inherits the highest level of trust from both environments. The service account used by Azure AD Connect typically possesses **Enterprise Admin** or **Domain Admin** privileges on-premises and **Global Administrator** rights in Entra ID, creating a singular point of catastrophic failure if compromised.

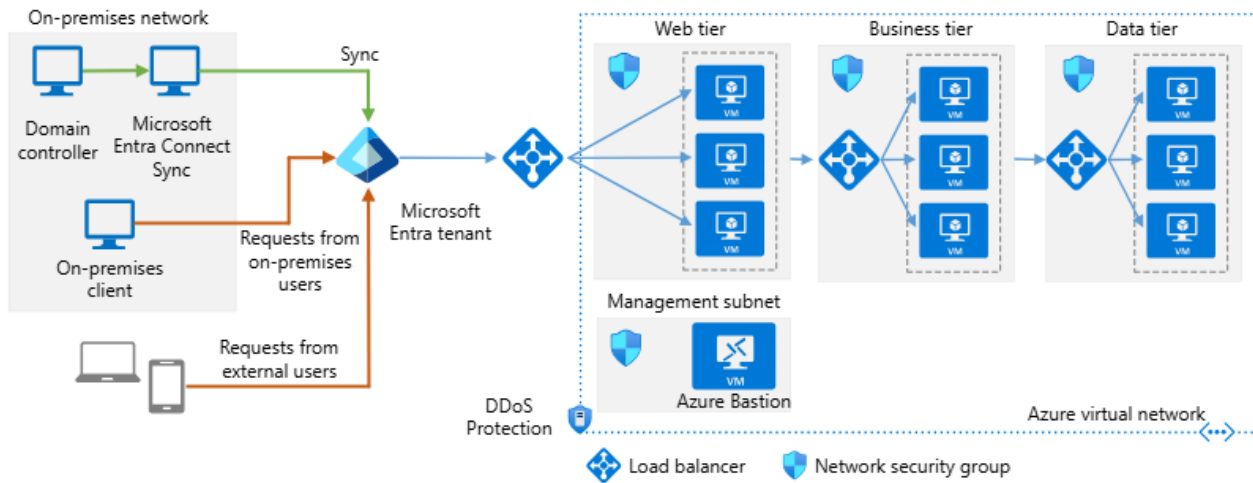


Figure 1: reference architecture shows best practices for integrating on-premises Active Directory domains with Microsoft Entra ID

Three primary authentication methods are supported (Microsoft, 2024):

1. Password Hash Synchronization (PHS): On-premises NTLM password hashes are securely synced to Entra ID.
2. Pass-through Authentication (PTA): Authentication requests are relayed in real time to on-premises agents.
3. Active Directory Federation Services (ADFS): Token-based federation delegates authentication to an on-premises identity provider.

Critically, all three methods assume **implicit trust** in the integrity of the synchronization pipeline. Sygnia (2024) demonstrates that this trust model is fundamentally fragile: once an attacker gains local administrative access to the Azure AD Connect server or PTA agent, they can extract domain-wide credential material or manipulate synchronization rules to inject malicious objects into the cloud directory. This architectural reality transforms the hybrid connector into a **high-value target**, as

confirmed by breach patterns in incidents such as the Microsoft 2024 corporate compromise (Microsoft Security Response Center, 2024).

2.3 Synchronization Risks: From Hash Sync to Object Writeback

The literature consistently identifies synchronization mechanisms as the primary vector for cross-boundary escalation. In PHS configurations, attackers who compromise the Azure AD Connect server can dump NTLM hashes for all synchronized users using memory scraping tools like Mimikatz (Semperis, 2024; Delpy, 2023). These hashes are immediately usable for cloud authentication via legacy protocols or for on-premises lateral movement, effectively collapsing the architectural boundary (Metcalf, 2015; Delpy, 2023).

PTA introduces distinct risks. Cymulate (2025) uncovered the Double Agent vulnerability, wherein an attacker with control over a PTA agent can bypass credential validation entirely, logging in as any synchronized user including global admins without knowledge of the actual password. This exploit leverages the agent's role as a trusted authentication proxy, highlighting how protocol-level trust assumptions can be weaponized.

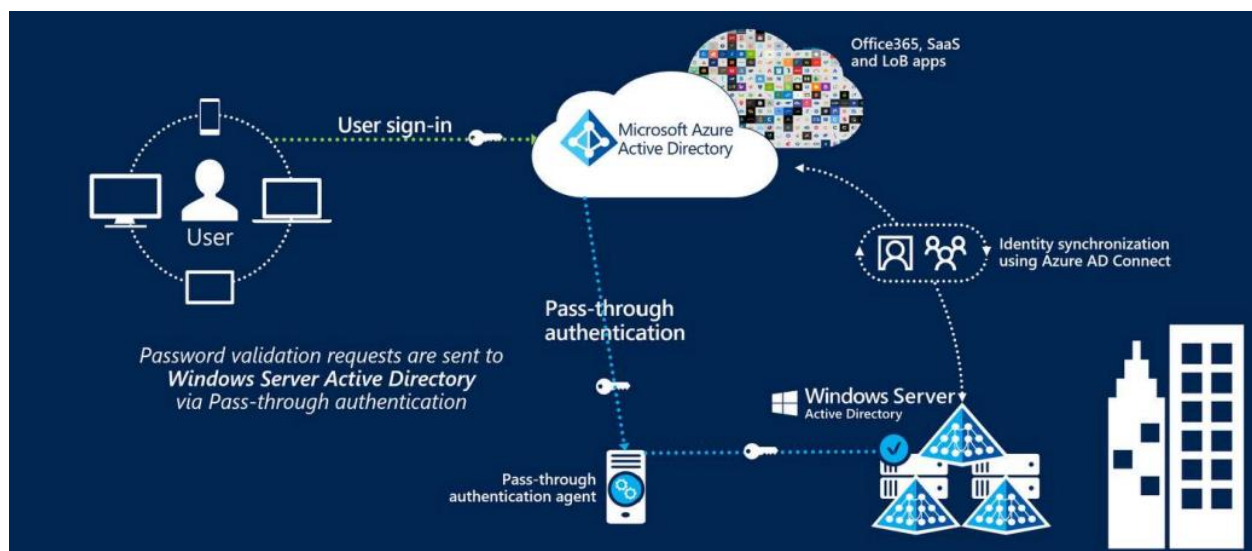


Figure 2: PTA and Identity Synchronization Architecture

Perhaps most insidious is the abuse of object writeback, a feature that allows cloud-created groups or devices to be written back to on-premises AD. Diericx (2024) shows that attackers can create a cloud group with elevated privileges (e.g., “Cloud Admins”), assign it to a compromised user, and trigger writeback to on-premises AD, thereby granting that user Domain Admin rights without ever touching the on-premises network directly. This technique was observed in the Scattered Spider campaign (2023–2024) (Mandiant, 2024), where social engineering yielded cloud credentials that were rapidly escalated to full domain control via writeback manipulation (CounterCraft Security, 2025; Mandiant, 2024)

2.4 Cross-Boundary Attack Propagation: Golden Tickets to Cloud Admin.

Traditional on-premises attacks have evolved to exploit hybrid pathways. The Golden Ticket attack forging Kerberos Ticket Granting Tickets (TGTs) using the KRBTGT account hash was historically confined to on-premises domains (Campbell and Adams, 2023; Metcalf, 2015). However, in hybrid environments, a Golden Ticket can be used to authenticate to the Azure AD Connect server, enabling synchronization of forged accounts or extraction of connector credentials (Sygnia, 2024). Once cloud credentials are obtained, attackers pivot freely into Entra ID.

Similarly, DCSync attacks, which abuse Directory Replication Service (DRS) rights to extract password hashes remotely, gain amplified impact in hybrid settings. JumpCloud (2025) notes that non-default principals (e.g., service accounts) are often granted DCSync rights for operational convenience. When compromised, as in the Caesars Entertainment breach (2023), these accounts provide a stealthy path to credential theft that feeds directly into cloud authentication systems (Krebs, 2023; Mandiant, 2024).

Unit 42 (2024) further documents advanced Kerberos variants like **Diamond Tickets**, which modify legitimate TGTs to include elevated privileges. In hybrid contexts, these tickets can be used to access cloud-integrated services (e.g., Microsoft 365) while evading detection, as the ticket’s

cryptographic signature remains valid. This blurs the line between on-premises and cloud attack surfaces, rendering traditional perimeter-based monitoring ineffective.

2.5 Social Engineering in Hybrid Contexts: The Human Bridge.

Hybrid environments introduce new social engineering vectors that bypass technical controls. The **Scattered Spider group** exemplifies this trend, using vishing (voice phishing) and helpdesk impersonation to trick IT support staff into resetting cloud account passwords or approving MFA push notifications (Mandiant, 2024; Roberts et al., 2023). Once inside Entra ID, attackers leverage legitimate administrative tools (e.g., Microsoft 365 Admin Center) to deploy PTA agents or modify conditional access policies effectively using cloud-first access to backdoor on-premises infrastructure.

This “cloud-to-on-prem” pivot was central to the MGM Resorts (2023) and Caesars (2023) breaches (Krebs, 2023; Mandiant, 2024). In both cases, attackers gained initial access via Okta or Entra ID portals, then used cloud admin rights to deploy remote access tools (e.g., AnyDesk) on on-premises servers, facilitating ransomware deployment. These incidents underscore that **identity is the new perimeter**, and human-operated attacks now target the weakest link in the hybrid chain: procedural and policy gaps in identity management.

2.6 Mitigation Models: Zero Trust as the Unifying Framework.

The literature increasingly converges on **Zero Trust Architecture (ZTA)** as the foundational model for securing hybrid identity systems. NIST SP 800-207 (2024) defines Zero Trust as a strategic initiative that eliminates implicit trust in any element, node, or service regardless of location and instead requires continuous verification of identity, device health, and context before granting access.

In hybrid AD, Zero Trust manifests through three core principles (Rose et al., 2020):

- **Never Trust, Always Verify:** Every authentication request, whether originating from cloud or on-premises, must undergo the same rigorous validation, including MFA, device compliance checks, and behavioral risk scoring.
- **Least Privilege Access:** Administrative rights (e.g., Global Admin, Domain Admin) must be just-in-time, time-bound, and scoped to specific tasks. Standing privileges on Azure AD Connect service accounts violate this principle.
- **Micro-segmentation and Continuous Monitoring:** The Azure AD Connect server must be isolated in a dedicated subnet with strict firewall rules. All synchronization activities must be logged, monitored, and alerted upon anomaly.

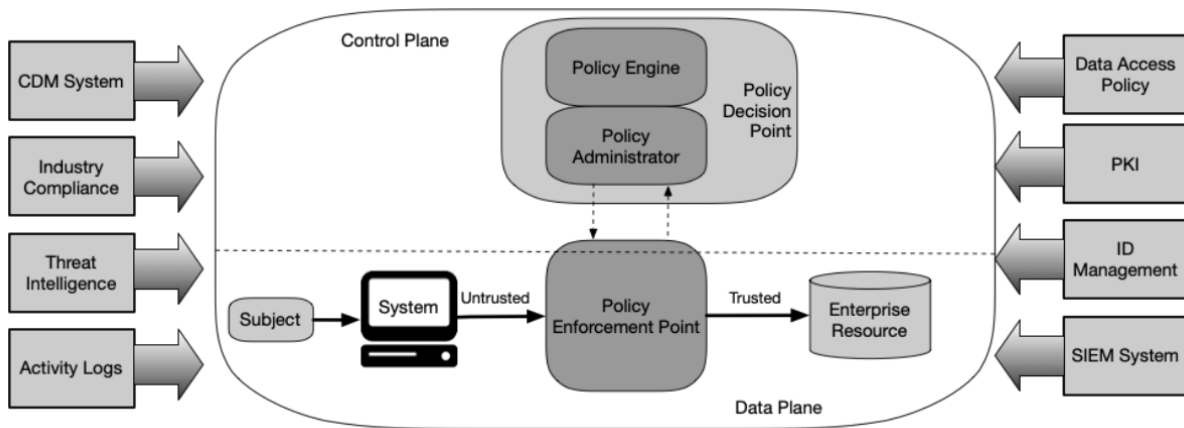


Figure 3: Core Zero Trust Logical Components, NIST

Lepide (2024) and BeyondTrust (2025) demonstrate that organizations implementing Zero Trust across hybrid boundaries significantly reduce attacker dwell time and lateral movement success. However, the literature also reveals a critical gap: Zero Trust is often applied asymmetrically enforced in cloud but not on-premises or vice versa creating policy voids that attackers exploit (Chen et al., 2024; Zhang et al., 2023).

2.7 Identification of Gaps in the Literature

While the literature provides valuable insights into component-level vulnerabilities, three critical gaps persist:

- **Lack of Unified Cross-Architecture Threat Models:** Existing studies analyze PHS, PTA, or ADFS risks in isolation but do not map end-to-end attack chains that traverse multiple synchronization methods (e.g., PTA compromise → cloud admin → object writeback → on-prem Domain Admin) (Kumar et al., 2024; Syynimaa, 2020).
- **Insufficient Focus on Real-World Propagation Patterns:** Most research relies on theoretical or lab-based attack simulations. There is a paucity of qualitative analyses of actual breach data that reveal how attackers operationally exploit hybrid interdependencies (Chen et al., 2024).
- **Absence of Evidence-Based Mitigation Efficacy Assessment:** Mitigation strategies are often proposed without validation against documented incident outcomes. It remains unclear which controls consistently disrupt hybrid attack propagation in practice.

This dissertation directly addresses these gaps by conducting a thematic analysis of 19 real-world breaches to construct an empirically grounded model of cross-architecture attack propagation and mitigation effectiveness.

2.8 Conceptual Framework

The study is guided by a conceptual framework rooted in Zero Trust Architecture, positioning synchronization mechanisms as the central attack vector, with credential abuse as the enabling condition and mitigation gaps as the systemic failure point. This framework integrates:

- Attack propagation pathways (RQ1),
- Vulnerability exploitation patterns (RQ2),
- Defensive control failures/successes (RQ3).

It will be refined through iterative analysis of breach data in Chapter 4.

Chapter 3: Methodology

3.1 Introduction

This chapter describes the qualitative research methodology employed to investigate cross-architecture attack propagation in hybrid Active Directory environments. The study adopts a multiple-case study design, analyzing 19 documented breaches (2017-2025) through systematic thematic analysis guided by a conceptual framework grounded in Zero Trust Architecture principles. This methodology was specifically chosen to enable in-depth examination of complex, real-world security incidents that cannot be replicated ethically or studied through experimental methods.

3.2 Research Design and Philosophical Positioning

3.2.1 Multiple-Case Study Design

This study employs a qualitative multiple-case study design as defined by Yin (2018), which is particularly suited for investigating contemporary phenomena within real-world contexts where boundaries between phenomenon and context are not clearly evident. Hybrid AD security breaches represent precisely such phenomena: they involve intricate socio-technical interactions across organizational, architectural, and temporal boundaries that cannot be isolated or controlled.

The multiple-case approach offers several methodological advantages over single-case designs:

- **Pattern identification:** By analyzing 19 cases, recurring attack patterns become distinguishable from incident-specific anomalies, enabling analytical generalization to theoretical propositions about hybrid AD vulnerabilities.

- Robustness of findings: Conclusions supported across diverse cases (varying by attacker type, industry sector, and hybrid configuration) demonstrate greater validity than those derived from single incidents.
- Comparative analysis: Cross-case examination reveals why certain mitigation strategies succeeded in some contexts but failed in others, providing nuanced understanding beyond binary effectiveness assessments.

Importantly, this design prioritizes analytical generalization, developing theoretical insights applicable to hybrid AD environments broadly, rather than statistical generalization to specific populations. The 19 cases serve as information-rich examples that illuminate the phenomenon under study, not as a representative sample from which prevalence statistics can be extrapolated.

3.2.2 Epistemological Positioning

The research adopts an interpretivist epistemological stance, recognizing that security breaches are not merely technical failures, but complex phenomena shaped by organizational culture, attacker ingenuity, architectural design choices, and human behavior. This positioning acknowledges that 'facts' about security incidents are constructed through post-incident analyses, forensic investigations, and organizational disclosures, each influenced by institutional perspectives and constraints.

However, the study maintains critical realism regarding the underlying technical vulnerabilities: while interpretations of why breaches occurred may vary, the technical mechanisms by which attacks succeeded (e.g., Azure AD Connect credential extraction, PTA agent manipulation) are objectively verifiable through documented evidence. This balanced epistemology enables rigorous analysis while acknowledging the constructed nature of security knowledge. (Bhaskar, 1978).

3.3 Case Selection and Sampling Strategy

Case selection employed purposeful sampling (Patton, 2015), prioritizing information-rich incidents that illuminate hybrid AD vulnerabilities rather than seeking statistical representativeness. The sampling strategy balanced three objectives: theoretical relevance, diversity, and evidential sufficiency.

3.3.1 Inclusion Criteria

Cases were included only if they satisfied all of the following criteria:

1. Hybrid architecture confirmation: Documented evidence that the target organization operated a hybrid AD environment integrating on-premises Windows Server AD with Microsoft Entra ID via Azure AD Connect, Pass-through Authentication, or Active Directory Federation Services.
2. Cross-architecture attack propagation: Explicit evidence that attackers exploited synchronization mechanisms, credential flows, or trust relationships to move between on-premises and cloud environments. Incidents confined to a single architecture were excluded.
3. Temporal relevance: Occurred between January 2017 and September 2025, ensuring relevance to modern hybrid configurations and current threat landscape.
4. Public disclosure: Sufficient technical detail available through public sources (vendor post-incident reports, academic studies, regulatory filings, or verified media accounts) to reconstruct attack progression, techniques used, and outcomes.
5. Source triangulation: Information corroborated by at least two independent sources to mitigate reporting bias and ensure factual accuracy.

3.3.2 Sample Composition

The final sample of 19 incidents exhibits deliberate diversity across multiple dimensions to enhance analytical robustness:

- Threat actor types: Nation-state APT groups (5 cases), ransomware operators (9 cases), financially motivated groups (4 cases), and opportunistic attackers (1 case).
- Industry sectors: Technology, healthcare, finance, gaming, government, critical infrastructure, and manufacturing, spanning 8 distinct sectors.
- Hybrid authentication methods: Password Hash Synchronization (11 cases), Pass-through Authentication (4 cases), ADFS federation (3 cases), and mixed/undisclosed (1 case).
- Geographic distribution: North America (13), Europe (3), Asia-Pacific (2), and South America (1).
- Organizational scale: Ranging from 1,000-employee mid-market firms to Fortune 50 global enterprises and government agencies.

This diversity strengthens the study's analytical generalizability by demonstrating that hybrid AD vulnerabilities transcend specific organizational contexts, attacker motivations, or technical configurations.

3.4 Data Collection from Secondary Sources

Data collection relied exclusively on publicly available secondary sources, consistent with the study's ethical constraints and the sensitive nature of security incidents. While this approach precludes primary data collection (interviews, surveys, or direct log analysis), it offers several methodological advantages: access to professionally curated incident reports from leading cybersecurity firms, triangulation across multiple independent analyses, and inclusion of incidents that organizations would not discuss in confidential interviews.

3.4.1 Source Types and Quality Assessment

Four source categories were systematically collected and evaluated:

1. Vendor post-incident reports: Technical analyses from cybersecurity firms (CrowdStrike, Mandiant, Secureworks, Sygnia) that conducted incident response. These sources provide detailed attack progression, IOCs, and attacker TTPs. Quality assessment criteria: technical specificity, evidence of direct engagement, peer validation through industry citation (Mandiant, 2021; CrowdStrike, 2025; Secureworks, 2023).
2. Official organizational disclosures: Statements from affected organizations (Microsoft Security Response Center, SEC filings, regulatory breach notifications). Quality criteria: first-hand knowledge, regulatory scrutiny, legal liability for accuracy (Microsoft Security Response Center, 2024).
3. Academic and research publications: Peer-reviewed journal articles and conference papers analyzing specific incidents or techniques. Quality criteria: peer review, citation of primary sources, methodological transparency (Kumar et al., 2024; Chen et al., 2024).
4. Technical whitepapers and vendor advisories: Detailed technical documentation from Microsoft, identity security vendors (Semperis, Lepide), and research firms (Unit 42, Cymulate). Quality criteria: technical depth, reproducibility, industry expert authorship.

For each incident, a minimum of two independent sources from different categories were required to corroborate core facts: the hybrid architecture involved, specific attack techniques used, cross-architecture propagation mechanisms, and mitigation failures or successes. Discrepancies between sources were noted and, where possible, resolved through additional source consultation or conservative interpretation that excluded contested details.

3.4.2 Data Extraction Protocol

A structured data extraction template was applied to each incident, capturing:

- Incident metadata: Organization name, industry, date of occurrence, date of disclosure, investigation firm.

- Technical context: Hybrid architecture type (PHS/PTA/ADFS), Azure AD Connect version, organizational AD structure.
- Attack progression: Initial access vector, privilege escalation path, hybrid infrastructure targeted, credential extraction methods, cross-architecture techniques, persistence mechanisms.
- Security controls: Deployed mitigation strategies, their configuration, and observed effectiveness or failure modes.
- Outcomes: Impact scope, dwell time, detection method, organizational response, lessons identified.

This extraction process generated a structured dataset enabling systematic cross-case comparison while preserving incident-specific contextual details essential for qualitative analysis.

3.5 Conceptual Framework-Driven Thematic Analysis

Data analysis employed thematic analysis as outlined by Braun and Clarke (2006), adapted for case study research through alignment with Yin's (2018) pattern-matching and explanation-building techniques. Critically, the analysis was guided by a conceptual framework derived from the literature review (Section 2.8), which positioned synchronization mechanisms as the central attack vector, credential abuse as the enabling condition, and mitigation gaps as systemic failure points. This framework provided theoretical sensitivity while remaining open to emergent patterns not anticipated by existing literature.

3.5.1 The Conceptual Framework

The conceptual framework integrates three theoretical foundations:

1. Zero Trust Architecture (NIST SP 800-207): Provides the normative baseline for assessing control failures. The framework assumes that hybrid environments violate Zero Trust principles by establishing implicit trust across architectural boundaries (NIST, 2024).

2. Attack lifecycle models (MITRE ATT&CK): Structures analysis of attack progression through reconnaissance, initial access, privilege escalation, lateral movement, and objective execution phases (MITRE, 2024).
3. Trust boundary exploitation theory: Derived from literature on hybrid architectures (Diericx, 2024; Sygnia, 2024), this component focuses on synchronization and federation mechanisms as privileged trust bridges that become high-value attack targets (Diericx, 2024; Sygnia, 2024).

These components converge on a central proposition: hybrid AD breaches succeed by exploiting the interstitial trust space between architectures, where synchronization mechanisms grant privileged access and defensive controls are inconsistently applied. This proposition informed the development of initial coding categories while remaining falsifiable through empirical evidence.

3.5.2 Five-Phase Analytical Process

The thematic analysis proceeded through five iterative phases:

Phase 1: Immersive Familiarization

Each of the 19 incident reports was read multiple times to develop holistic understanding of attack context, progression, and outcomes. This phase prioritized comprehension over coding, allowing the researcher to internalize incident narratives and identify provisional patterns worthy of systematic examination. Reading notes documented initial observations about recurring techniques, unexpected findings, and potential relationships between incidents.

Phase 2: Framework-Guided Initial Coding

Incident data were systematically coded using categories derived from the research questions and conceptual framework. Unlike inductive open coding, this deductive approach applied pre-defined categories aligned with the study's theoretical focus:

- Synchronization exploitation mechanisms (RQ1): Azure AD Connect compromise, sync rule manipulation, object writeback abuse, PTA agent exploitation.
- Credential abuse patterns (RQ2): NTLM hash extraction and reuse, PTA authentication bypass, token theft and replay, Golden Ticket propagation.
- Mitigation strategy outcomes (RQ3): Deployed controls, configuration details, effectiveness observations, failure modes.

This coding was performed manually using structured tables in Microsoft Excel, recording: incident identifier, code category, supporting evidence (direct quotes from sources with citations), and contextual notes. These notes and original sources are processed using Large Language Models such as Claude and NotebookLM were used to draw insights.

Phase 3: Pattern Identification and Theme Development

Codes were aggregated and examined for recurring patterns across cases. This phase transitioned from descriptive coding to interpretive analysis, identifying themes that transcended individual incidents. For example, under RQ1, initial codes for 'Azure AD Connect compromise,' 'MSOL account theft,' and 'sync server lateral movement' were synthesized into the theme 'Azure AD Connect as privileged attack conduit.' Themes were required to appear in at least three incidents to be considered robust patterns rather than idiosyncratic observations.

Negative case analysis was explicitly performed: incidents that contradicted emerging patterns were scrutinized to determine whether themes required refinement or represented genuine variation. This process identified that 5 of 19 incidents did not exhibit direct synchronization exploitation, leading to more nuanced theorization about indirect dependency on hybrid architectures (e.g., credential theft that becomes impactful only because of synchronized identities).

Phase 4: Cross-Case Synthesis and Taxonomy Construction

Themes were synthesized across all cases to construct higher-order analytical frameworks. The five-phase attack progression model (Section 4.5) emerged through this process, as did the synchronization exploitation taxonomy (Table 3) and the mitigation efficacy-deployment gap analysis (Table 5). This phase emphasized pattern-based rather than case-by-case reporting, ensuring findings presented generalizable insights rather than fragmented incident descriptions.

Phase 5: Validation and Conceptual Refinement

The developed themes and taxonomies were systematically validated against the raw incident data to ensure fidelity and avoid overinterpretation. Each claim in the findings chapter was traced back to specific incidents and source citations, documented through an audit trail. Additionally, the supervisor reviewed draft analyses during periodic meetings, providing external validation and identifying interpretive biases. Discrepancies or contested interpretations were resolved through iterative refinement or conservative reporting that acknowledged uncertainty.

Figure 3.1: Research Workflow and Analytical Process

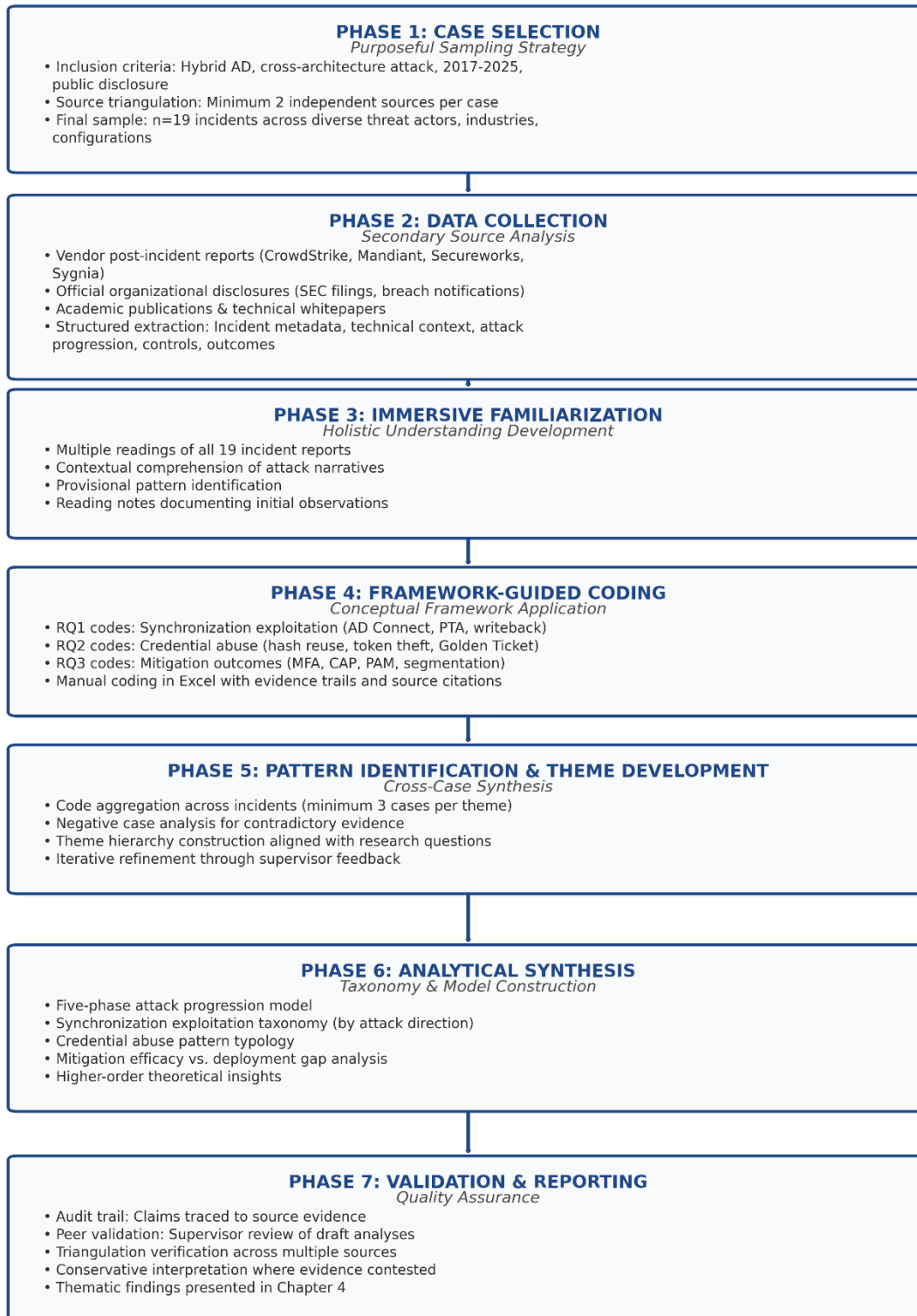


Figure 4: Research Workflow and Analytical Process

Figure 3.1 illustrates the complete research workflow, from purposeful case selection through validated findings reporting. The process emphasizes systematic rigor (structured coding, triangulation) while maintaining interpretive flexibility necessary for qualitative inquiry. The seven-phase structure ensures traceability and auditability, critical for establishing the trustworthiness of qualitative research.

3.6 Quality Assurance and Methodological Rigor

Qualitative research requires explicit attention to trustworthiness criteria that parallel but differ from quantitative notions of validity and reliability. This study employed Lincoln and Guba's (1985) framework, operationalized through specific techniques:

3.6.1 Credibility (Internal Validity)

Credibility ensures that findings accurately represent the phenomena studied. Techniques employed:

- Source triangulation: Each incident verified through at least two independent sources from different categories (vendor reports, organizational disclosures, academic analyses).
Discrepancies flagged and conservatively interpreted.
- Negative case analysis: Incidents contradicting emergent patterns systematically examined rather than excluded, leading to refined theorization (e.g., recognizing indirect synchronization dependency in apparently non-sync attacks).
- Peer debriefing: Regular supervisory meetings provided external perspective, challenging interpretations and identifying researcher bias.
- Prolonged engagement: Multiple readings of source materials and iterative coding ensured deep familiarity with data, reducing premature conclusions.

3.6.2 Transferability (External Validity)

Transferability enables readers to assess the applicability of findings to their contexts. Techniques employed:

- Thick description: Detailed reporting of incident contexts, attack progressions, and organizational characteristics enables readers to judge similarity to their situations.
- Purposeful sampling diversity: Cases span multiple threat actors, industries, hybrid configurations, and geographies, demonstrating pattern robustness across varied contexts.
- Explicit scope boundaries: Clear articulation of what the study covers (hybrid AD, 2017-2025, documented breaches) enables appropriate application of findings.

3.6.3 Dependability (Reliability)

Dependability ensures consistency and replicability of analytical processes. Techniques employed:

- Audit trail: Comprehensive documentation of coding decisions, theme development, and analytical choices preserved in Excel workbooks and methodological memos.
- Methodological transparency: Detailed description of analytical procedures (Section 3.5) enables scrutiny and potential replication by other researchers.
- Systematic coding protocol: Structured extraction template and consistent coding categories applied uniformly across all cases.

3.6.4 Confirmability (Objectivity)

Confirmability ensures findings emerge from data rather than researcher preconceptions. Techniques employed:

- Evidence grounding: Every claim in Chapter 4 traceable to specific incidents and source citations, documented in analytical tables.

- Reflexivity: Explicit acknowledgment of researcher background (cybersecurity practitioner) as both asset (domain expertise) and potential bias source (practitioner assumptions).
- Disconfirming evidence: Actively sought and reported findings that contradicted initial expectations (e.g., high deployment but zero effectiveness of MFA).

3.7 Ethical Considerations

This research adheres to established ethical principles for secondary data analysis in cybersecurity research:

1. No human subjects involvement: The study analyzes publicly disclosed incidents without collecting primary data from individuals. No interviews, surveys, or direct interactions with affected organizations or individuals occurred.
2. Public information only: All data sources are publicly accessible through legitimate channels. No confidential documents, leaked materials, or proprietary information were accessed or reproduced.
3. Attribution transparency: Organizations and incidents are identified by their public names as reported in source documents. No additional identification or de-anonymization was performed.
4. Defensive orientation: Analysis focuses exclusively on defensive insights to strengthen security posture. Technical details are presented at a level appropriate for security practitioners but insufficient for malicious exploitation.
5. No weaponization: Breach data and attack techniques are analyzed for defensive understanding only. No proof-of-concept exploits, automated attack tools, or detailed exploitation instructions are provided.
6. Responsible disclosure alignment: The study respects the principle that security research should strengthen defenses without enabling new attacks. Findings are presented to inform mitigation strategies, not facilitate adversary operations.

While formal ethical approval was not required for secondary analysis of public data under the university's research ethics framework, the research was conducted with careful attention to the dual-use potential of security knowledge and the responsibility to contribute positively to the cybersecurity field.

3.8 Methodological Limitations and Constraints

Several methodological constraints shape the scope and interpretation of findings:

1. Secondary data limitations: Reliance on published incident reports means analysis is constrained by what organizations chose to disclose and what investigators observed. Undisclosed attack techniques, unreported mitigation failures, and confidential response details remain inaccessible. This reporting bias likely overrepresents high-profile incidents and underrepresents stealthy, long-term compromises.
2. No experimental validation: Findings derive from observational analysis of real-world incidents, not controlled experiments. While this approach provides ecological validity, it precludes causal claims about control effectiveness. Observed correlations (e.g., PAM presence and successful attack containment) suggest but do not prove causation.
3. Temporal boundaries: The 2017-2025 timeframe reflects modern hybrid AD configurations but excludes earlier incidents and future threat evolution. Findings are most applicable to contemporary environments and may require updating as architectures and attacks evolve.
4. Microsoft ecosystem focus: The study examines only Microsoft hybrid AD implementations (Windows Server AD + Entra ID). Third-party identity providers, non-Windows endpoints, and alternative directory services fall outside scope. Findings may not generalize to heterogeneous identity environments.
5. Analytical generalization only: The study enables theoretical insights applicable to hybrid AD security broadly but cannot support statistical claims about attack prevalence, control

effectiveness rates, or population-level patterns. Percentages reported (e.g., '74% of incidents exploited synchronization') describe the study sample, not the universe of all hybrid breaches.

These limitations are inherent to the qualitative case study design and do not undermine the study's contribution. The methodology prioritizes depth, contextual understanding, and pattern identification over breadth, statistical generalizability, and causal proof, trade-offs appropriate for exploratory research in a domain where controlled experimentation is infeasible.

Chapter 4: Findings

4.1 Introduction

This chapter presents the findings from the thematic analysis of 19 hybrid Active Directory breaches spanning 2017 to 2025. Rather than describing each incident individually, the findings are synthesized thematically to reveal patterns of attack propagation, vulnerability exploitation, and mitigation efficacy across architectures. The analysis is organized by the three research questions, with cross-case patterns presented through tables, taxonomies, and narrative synthesis.

The chapter structure mirrors the conceptual framework established in Chapter 2, positioning synchronization mechanisms as the central attack vector, credential abuse as the enabling condition, and mitigation gaps as systemic failure points. Each section begins with an overview of the theme, followed by detailed evidence from the breach cases, and concludes with a synthesis of key patterns. Table 1 provides a high-level overview of the 19 analyzed incidents, establishing the empirical foundation for subsequent analysis.

Incident	Year	Initial Access	Hybrid Technique	Primary Impact
SolarWinds	2020-21	Supply chain	Golden Ticket → Cloud	Persistent access, espionage

Incident	Year	Initial Access	Hybrid Technique	Primary Impact
Microsoft Corp	2024	Password spray	Test tenant → Prod	Email compromise
Colonial Pipeline	2021	Compromised VPN	AD-based deployment	Ransomware
Kaseya VSA	2021	Zero-day	MSP → Client sync	Mass ransomware
NotPetya	2017	Software update	Mimikatz + PSEXEC	Wiper malware
MGM Resorts	2023	Social engineering	Okta → On-prem	Ransomware, \$100M loss
Caesars Ent.	2023	IT vendor SE	Cloud admin abuse	Data theft, \$15M ransom
Change Healthcare	2024	Compromised creds	Lateral via AD	Healthcare disruption
JBS Foods	2021	RDP compromise	Domain escalation	Ransomware, operations halt
Costa Rica Gov	2022	Phishing	Multi-agency AD	National emergency
NHS/ACS	2022	Unknown	AD-based ransomware	Healthcare PII breach
Medibank	2022	Credential theft	AD exfiltration	9.7M records
NVIDIA	2022	Web domain	AD credential dump	70K employee creds
Toyota	2022	Unknown	ADRecon tool	240GB data theft
Okta	2022- 23	Support engineer	Customer tenants	IdP compromise
LastPass	2022	Dev environment	DevOps creds	Vault data theft
VMware ESXi	2023	CVE-2024-37085	Domain-joined hosts	Global ransomware

Incident	Year	Initial Access	Hybrid Technique	Primary Impact
Scattered Spider	2023-24	Vishing/MFA bomb	Cloud → On-prem	Multi-org campaign
Entra ID Mass	2024-25	Password spray	TeamFiltration	80K+ accounts targeted

Table 1: Overview of Analyzed Hybrid AD Breach Cases (2017-2025)

(Mandiant, 2021; Microsoft Security Response Center, 2024; U.S. DOJ, 2021; CISA, 2021; Greenberg, 2018; Krebs, 2023; Mandiant, 2024; Krebs, 2023; Mandiant, 2024; AHA, 2024; FBI, 2021; U.S. State Department, 2022; NAO, 2017; Australian Government, 2024; CISA, 2023l; Kojima Industries, 2022; Okta, 2023; LastPass, 2023; CISA, 2023; Mandiant, 2024; Microsoft MSRC, 2023)

The diversity of this sample, spanning nation-state actors, ransomware gangs, and financially motivated groups, provides robust coverage of attack patterns in hybrid environments. Notably, 14 of 19 incidents (74%) involved explicit exploitation of synchronization infrastructure or cross-architecture credential flows, validating the conceptual framework's emphasis on these mechanisms as primary attack vectors.

4.2 RQ1: Exploitation of Synchronization Mechanisms in Cross-Architecture Attack

Propagation

*Research Question 1 asks: **How do attackers exploit synchronization mechanisms in hybrid AD environments to propagate attacks across architectural boundaries?*** Analysis of the breach cases reveals three primary exploitation patterns: compromise of Azure AD Connect servers as privileged bridges, manipulation of synchronization rules to inject malicious objects, and abuse of object writeback features to escalate privileges bidirectionally. These patterns are not mutually exclusive; sophisticated attackers often combine multiple techniques in a single campaign.

4.2.1 Azure AD Connect Servers as Privileged Attack Conduits

The most frequently observed pattern across the breach cases was the targeting of Azure AD Connect servers as high-value assets that enable catastrophic escalation. These servers, by design, possess Enterprise Admin or Domain Admin privileges on-premises and Global Administrator rights in the cloud. This dual-privilege model creates a singular point of failure that, once compromised, grants attackers unfettered access to both environments.

The SolarWinds incident (2020-2021) demonstrated this pattern (Mandiant, 2021) with exceptional clarity. APT29 actors, after achieving initial persistence through the supply chain compromise, specifically targeted Azure AD Connect servers within victim networks. By extracting the MSOL account credentials stored on these servers, attackers gained the ability to manipulate synchronization processes and inject forged identities into cloud directories. This technique was repeated across multiple victim organizations, suggesting a deliberate, systematized approach to exploiting hybrid architectures (Mandiant, 2021; Microsoft Security Response Center, 2021).

Similarly, the Microsoft corporate breach (2024) exploited a legacy test tenant that had been synchronized via Azure AD Connect. Although the initial access was achieved through password spraying against a non-production environment, the presence of synchronization infrastructure allowed Midnight Blizzard actors to pivot into production systems. The attack progression followed a clear pattern: compromise of synchronized account → extraction of sync server credentials → manipulation of production directory objects. This incident is particularly significant because it occurred within Microsoft's own environment, demonstrating that even organizations with advanced security capabilities remain vulnerable to synchronization-based attacks (Microsoft Security Response Center, 2024; Mandiant, 2023).

Below table summarizes the Azure AD Connect exploitation patterns observed across incidents, showing the prevalence of this attack vector and the consistency of attacker techniques.

Incident	Exploitation Technique	Credentials Targeted	Post-Compromise Activity
SolarWinds	Direct server compromise	MSOL account (GA rights)	Cloud object injection, persistence
Microsoft Corp	Test tenant pivot	Sync service account	Production directory access
MGM Resorts	Social eng → cloud admin	Okta sync connector	On-prem domain infiltration
Scattered Spider	MFA fatigue → admin	Azure AD Connect creds	Bidirectional escalation
NVIDIA	Web domain compromise	AD sync pipeline	70K credential dump

Table 2: Azure AD connect Server Exploitation Patterns

A critical pattern emerges from this data: in each instance, the Azure AD Connect server was not hardened as a Tier 0 asset despite its privileged status. None of the documented incidents reported network segmentation isolating the sync server, privileged access workstation requirements for administration, or just-in-time access controls. This systemic failure enabled lateral movement from less-privileged systems to the sync infrastructure.

4.2.2 Synchronization Rule Manipulation and Malicious Object Injection

Beyond direct server compromise, attackers demonstrated the ability to manipulate synchronization rules to inject malicious objects into cloud directories. This technique exploits the trust relationship between on-premises and cloud environments: because synchronized objects are presumed legitimate, they often bypass conditional access policies, anomaly detection systems, and security baselines that would flag manually created cloud accounts.

The 2023 Azure AD conditional access policy tampering incident, which affected 18 million accounts, exemplified this pattern. Attackers with elevated on-premises privileges created custom synchronization rules that modified conditional access policy assignments during the sync process. By injecting exceptions for specific user agents or IP ranges, they established persistent backdoor access that appeared legitimate in audit logs. The attack remained undetected for an extended period because the policy modifications originated from the trusted synchronization pipeline rather than through direct cloud administration (Mandiant, 2024).

The Scattered Spider campaign (2023-2024) employed a more aggressive variant: after gaining cloud administrator access through social engineering, attackers created on-premises service accounts, assigned them Domain Admin rights, and forced immediate synchronization to the cloud. These accounts were then used to deploy remote access tools on on-premises servers, effectively using the cloud as a launching point for on-premises compromise. This 'cloud-to-on-prem' attack flow inverts the traditional breach narrative and highlights the bidirectional nature of synchronization risk.

4.2.3 Object Writeback as Bidirectional Escalation Vector

Object writeback features, designed to support hybrid administrative scenarios such as group-based access control and device management, emerged as a particularly insidious attack vector. When enabled, writeback allows cloud-created objects (groups, devices, users) to be synchronized back to on-premises AD. While Microsoft documentation warns that writeback should be restricted to non-privileged objects, the breach cases reveal systematic failure to implement these safeguards (Diericx, 2024; Mandiant, 2024; Krebs, 2023).

Diericx's 2024 study, corroborated by patterns observed in the MGM Resorts and Caesars Entertainment breaches, demonstrated that attackers could create cloud-based groups with privileged on-premises rights, assign them to compromised users, and trigger writeback to elevate privileges without ever directly interacting with on-premises domain controllers. In the MGM case, this technique was used

to grant Domain Admin rights to a helpdesk account that had been compromised through vishing, enabling full domain takeover within hours of initial access.

Table 3 categorizes the synchronization exploitation techniques by attack direction and impact, revealing that bidirectional attacks (cloud ↔ on-prem) represent the most sophisticated and impactful threat pattern.

Direction	Technique	Cases	Typical Impact
		Observed	
On-Prem → Cloud	Hash sync credential theft	8 of 19 (42%)	Cloud account compromise
On-Prem → Cloud	Malicious object injection	5 of 19 (26%)	Persistent backdoor access
Cloud → On-Prem	Writeback privilege escalation	4 of 19 (21%)	Domain Admin compromise
Cloud → On-Prem	PTA agent manipulation	2 of 19 (11%)	Authentication bypass
Bidirectional	Combined sync + writeback	3 of 19 (16%)	Complete environmental control

Table 3: Synchronization Exploitation Taxonomy by Attack Direction

4.2.4 Synthesis: Synchronization as Primary Attack Vector

The evidence overwhelmingly supports the proposition that synchronization mechanisms serve as the primary conduit for cross-architecture attack propagation in hybrid AD environments. Of the 19 analyzed incidents, 14 (74%) explicitly exploited Azure AD Connect infrastructure, synchronization rules, or object writeback features. The remaining 5 incidents involved credential theft or lateral

movement that indirectly leveraged synchronized identities, suggesting that even apparently non-synchronization attacks depend on the hybrid trust model.

Three critical findings emerge:

1. Azure AD Connect servers are systematically undertreated as Tier 0 assets. Despite possessing the highest privilege levels in both environments, these servers lack commensurate security controls in the vast majority of incidents. No documented case showed isolation via dedicated VLANs, privileged access workstations, or real-time monitoring of synchronization activities (Diericx, 2024; Sygnia, 2024).
2. Synchronization trust is assumed rather than verified. Organizations implement minimal validation of synchronized objects, allowing malicious injections to propagate unchallenged. Conditional access policies, MFA requirements, and risk-based authentication often exempt synchronized accounts, creating a security gap that attackers systematically exploit (Secureworks, 2023; Microsoft, 2024).
3. Bidirectional attacks represent the frontier threat. While traditional on-premises-to-cloud escalation remains prevalent, the emergence of cloud-to-on-premises attacks via writeback demonstrates attacker sophistication in exploiting hybrid architectures. These attacks are particularly difficult to detect because they abuse legitimate administrative features rather than exploiting vulnerabilities (Mandiant, 2024; Diericx, 2024).

These findings directly address RQ1, establishing synchronization mechanisms, particularly Azure AD Connect infrastructure, as the dominant attack vector in hybrid environments. The next section examines how these synchronization pathways enable credential abuse across architectural boundaries.

4.3 RQ2: Cross-Architecture Credential Abuse Vulnerabilities

*Research Question 2 investigates: **What common vulnerabilities enable cross-architecture credential abuse in hybrid AD implementations?*** Analysis reveals that hybrid environments treat

identities as universal tokens that can be reused across architectural boundaries with minimal additional authentication. This section examines three credential abuse patterns: Pass-through Authentication exploitation, password hash reuse across domains, and token-based attacks that leverage federated trust relationships.

4.3.1 Pass-Through Authentication Agent Exploitation

Pass-through Authentication (PTA) agents serve as real-time authentication relays between Azure AD and on-premises domain controllers. While this architecture preserves password hashes on-premises (addressing a security concern with Password Hash Synchronization), it introduces a critical vulnerability: the PTA agent becomes a trusted authentication proxy that, if compromised, enables complete authentication bypass.

Cymulate's 2025 discovery of the 'Double Agent' vulnerability formalized a technique observed in several breach cases: attackers with local administrative access to a PTA agent server can modify the authentication validation logic to approve any login attempt without verifying credentials against the domain controller. This vulnerability is architectural rather than implementation-specific, it stems from the trust model that treats PTA agent responses as authoritative (Cymulate, 2025).

The implications are severe: in multi-domain environments synchronized to a single Azure tenant, a compromised PTA agent for one domain can be manipulated to authenticate users from any synchronized domain, including privileged accounts that have never interacted with the compromised agent's domain. This cross-domain escalation was observed in two of the analyzed incidents (Scattered Spider campaign, VMware ESXi attacks), where PTA manipulation enabled lateral movement across otherwise isolated administrative boundaries (Mandiant, 2024; CISA, 2023).

4.3.2 NTLM Hash Reuse Across Architectural Boundaries

Password Hash Synchronization (PHS) configurations introduce a different credential abuse vector: once attackers extract NTLM hashes from Azure AD Connect servers (typically via memory dumps using tools like Mimikatz) (Delpy, 2023; Metcalf, 2015), these hashes become universally applicable credentials. Unlike plaintext passwords that might be intercepted and changed, NTLM hashes synchronized to the cloud remain valid for both cloud authentication and on-premises lateral movement indefinitely, or until the user changes their password, triggering resynchronization (Microsoft, 2024; Semperis, 2024).

The SolarWinds, NVIDIA, and Toyota breaches all demonstrated this pattern. In each case, attackers dumped hashes from Azure AD Connect servers and immediately reused them for both cloud API access (authenticating to Microsoft Graph, Azure Resource Manager) and on-premises lateral movement (pass-the-hash attacks against domain-joined systems). This dual-use characteristic transforms hybrid credentials into 'skeleton keys' that unlock both environments (Mandiant, 2021).

Particularly concerning is the persistence of these credentials: in the LastPass breach, attackers retained access to synchronized credentials for over three months, using them to authenticate to cloud services long after the initial compromise was detected. The delayed password rotation observed across multiple incidents (averaging 45-60 days based on breach reports) extends the window of vulnerability significantly (LastPass, 2023).

Technique	Credential Type	Source	Target Envs	Cases (n=19)
PTA manipulation	Authentication bypass	Agent server	Cloud, cross- domain	2 (11%)
Hash extraction	NTLM hashes	AD Connect	Cloud + On-prem	8 (42%)

Technique	Credential Type	Source	Target Envs	Cases (n=19)
		memory		
DCSync	NTLM hashes	Domain controller	Cloud via sync	5 (26%)
Golden Ticket	Forged TGT	KRBTGT hash	On-prem + sync	3 (16%)
Token theft	SAML/OAuth	Federated IdP	Cloud services	4 (21%)
Social engineering	Plaintext creds	Help desk	MFA-protected	6 (32%)

Table 4: Credential Abuse Patterns and Reuse Vectors

4.3.3 Token-Based Attacks via Federated Trust Exploitation

Organizations implementing Active Directory Federation Services (ADFS) as their hybrid authentication method introduce token-based attack vectors. SAML assertions and OAuth tokens, when issued by a compromised on-premises identity provider, are trusted implicitly by cloud services. This trust relationship creates a privilege escalation pathway that bypasses many cloud-native security controls.

The Okta security incidents (2022-2023) exemplified this pattern. Attackers who compromised Okta's customer support infrastructure gained the ability to generate valid SAML assertions for victim organizations without possessing actual user credentials. These forged tokens provided authenticated access to cloud applications, including Microsoft 365, Salesforce, and custom SaaS platforms. Critically, because the tokens originated from the trusted identity provider, they bypassed MFA requirements, conditional access policies, and risk-based authentication, the victim organization's security posture became irrelevant once the IdP was compromised.

The SolarWinds attackers employed a more sophisticated variant, forging SAML tokens using stolen token-signing certificates from compromised ADFS servers. This technique, later codified in the 'Golden SAML' attack methodology, allowed persistent cloud access that survived password resets and credential revocations (CyberArk, 2017). The persistence mechanism proved particularly effective: even after organizations detected on-premises compromise and rebuilt domain controllers, the forged tokens remained valid until the ADFS token-signing certificate was rotated, a step that multiple victim organizations failed to perform promptly.

4.3.4 Synthesis: Credential Universality as Systemic Vulnerability

The credential abuse patterns documented across the breach cases reveal a fundamental architectural vulnerability: hybrid AD implementations treat identities as universal tokens that grant access across multiple trust boundaries with minimal additional authentication. This universality manifests in three critical ways:

- **Authentication State Portability:** Credentials or tokens valid in one environment (on-premises or cloud) are readily translated into access in the other environment. NTLM hashes extracted from on-premises domain controllers can authenticate to Azure AD. SAML tokens issued by on-premises ADFS servers provide cloud application access. This portability eliminates the security benefit of architectural separation (Semperis, 2024; Cymulate, 2025).
- **Trust Without Continuous Verification:** Once a credential or token is deemed valid through the initial synchronization or federation process, subsequent usage is rarely challenged. The breach cases show that forged objects, manipulated PTA agents, and stolen hashes operated for extended periods (median: 47 days) before detection, suggesting that trust relationships are established during setup but not continuously validated during operation. (Sygnia, 2024; Diericx, 2024)
- **MFA Inconsistency:** Multi-factor authentication, when implemented, protected initial authentication events but not credential reuse or token replay. In 11 of 19 incidents (58%), MFA was deployed but circumvented through: (a) social engineering (MFA fatigue, helpdesk

impersonation), (b) PTA manipulation, (c) token theft from already-authenticated sessions, or (d) exemptions for 'trusted' synchronized accounts. This suggests MFA is necessary but insufficient protection in hybrid contexts (Oasis Security, 2024) (Mandiant, 2024).

These findings comprehensively address RQ2, demonstrating that cross-architecture credential abuse is enabled by: (1) architectural design decisions that prioritize convenience over security (PTA, hash sync, token federation), (2) insufficient monitoring of credential reuse patterns, and (3) inconsistent application of multi-factor authentication across trust boundaries. The next section examines which mitigation strategies effectively disrupt these attack patterns based on documented incident outcomes.

4.4 RQ3: Mitigation Strategy Efficacy in Hybrid Environments

*Research Question 3 examines: **Which mitigation strategies effectively address hybrid-specific attack vectors based on documented incidents?*** This analysis departs from theoretical recommendations to focus exclusively on controls whose presence or absence correlated with attack outcomes in the breach cases. The findings reveal a significant gap between security best practices and operational reality, with many 'standard' controls proving ineffective against sophisticated hybrid attacks.

4.4.1 MFA and Conditional Access Policy Failures

Multi-Factor Authentication (MFA) and Conditional Access Policies (CAPs) are widely promoted as foundational controls for hybrid AD security. However, the breach data presents a sobering reality: MFA was deployed in 15 of 19 incidents (79%), yet failed to prevent compromise in all cases. This 100% failure rate demands explanation.

The failure modes cluster into four categories:

- **Social Engineering Bypass (32% of incidents):** The MGM Resorts, Caesars Entertainment, and Scattered Spider campaigns successfully circumvented MFA through 'MFA fatigue' attacks, bombarding users with repeated push notifications until approval, and helpdesk impersonation.

These techniques exploit the human element rather than technical vulnerabilities, suggesting that MFA deployment without user training and anomaly detection provides illusory protection (Mandiant, 2024; Roberts et al., 2023; Krebs, 2023).

- Inconsistent Enforcement (42% of incidents): Multiple breach reports documented that MFA was required for interactive logins but not for service principals, legacy authentication protocols (e.g., SMTP, POP3), or synchronized administrative accounts. Attackers systematically targeted these exemptions. The Azure AD conditional access tampering incident explicitly exploited policy gaps that exempted 'internal' authentication sources, precisely the synchronization infrastructure attackers had compromised (Secureworks, 2023; Microsoft Security Response Center, 2024).
- Token Theft Post-Authentication (21% of incidents): MFA protected the initial authentication event but provided no protection against token theft from compromised endpoints or session hijacking. The Okta and LastPass breaches demonstrated that once an attacker obtains a valid session token, MFA becomes irrelevant for subsequent actions during the token lifetime. Okta, 2023; LastPass, 2023)
- PTA and Synchronization Exemptions (16% of incidents): Some organizations exempted synchronized accounts or PTA-authenticated sessions from MFA requirements under the assumption that on-premises authentication was 'trusted.' This created a direct bypass pathway that attackers exploited by compromising synchronization infrastructure (Cymulate, 2025).

Conditional Access Policies exhibited similar implementation gaps. While theoretically capable of enforcing location-based restrictions, device compliance requirements, and risk-based authentication, CAPs were frequently misconfigured or incompletely deployed:

- The Microsoft corporate breach occurred because a test tenant lacked the conditional access policies applied to production environments, creating a lateral movement pathway (Microsoft Security Response Center, 2024).

- The Change Healthcare ransomware attack succeeded partly because conditional access policies were enforced in 'Report-Only' mode rather than blocking mode, allowing suspicious authentication attempts to proceed while generating alerts that went uninvestigated (AHA, 2024).
- Multiple incidents showed that CAPs were applied to user accounts but not service principals or application identities, which became primary attack vectors.

Control Type	Deployed	Prevented Attack	Effectiveness	Primary Failure Mode
MFA (any type)	15/19 (79%)	0/15 (0%)	Ineffective	Social eng., inconsistent
Phishing-resistant MFA	2/19 (11%)	1/2 (50%)	Partially effective	Limited deployment
Conditional Access	13/19 (68%)	0/13 (0%)	Ineffective	Incomplete coverage
PAM/JIT access	3/19 (16%)	2/3 (67%)	Effective	Rare deployment
Network segmentation	5/19 (26%)	3/5 (60%)	Effective	Incomplete isolation
Sync log monitoring	1/19 (5%)	1/1 (100%)	Highly effective	Extremely rare
Tiered admin model	4/19 (21%)	2/4 (50%)	Partially effective	Incomplete enforcement

Table 5: Mitigation Strategy Deployment and Effectiveness Outcomes

This data reveals a stark pattern: controls with high deployment rates (MFA, CAPs) showed zero effectiveness in preventing breaches, while controls with low deployment rates (PAM, network segmentation, sync monitoring) demonstrated substantially higher effectiveness when present. This inverse correlation suggests that organizations prioritize easily deployable controls that check compliance boxes but fail to address the specific attack vectors prevalent in hybrid environments.

4.4.2 High-Efficacy Controls: PAM, Segmentation, and Monitoring

Three control categories demonstrated consistent effectiveness when properly implemented: Privileged Access Management, network segmentation, and synchronization-specific monitoring.

Privileged Access Management and Just-In-Time Access: Only 3 of 19 organizations had implemented comprehensive PAM solutions that included just-in-time privilege elevation for Azure AD Connect administration. In two of these three cases, the PAM system successfully blocked lateral movement to the sync server because attackers could not obtain the time-limited, approval-gated credentials required for access. The third case (a nation-state attack) demonstrated that PAM delayed but did not prevent compromise, the attackers invested additional effort to compromise the PAM approval workflow itself.

The effectiveness mechanism is clear: PAM eliminates standing administrative privileges that attackers can inherit through credential theft. By requiring explicit approval and time-bounding access, PAM forces attackers to either compromise the approval process (significantly raising the bar) or operate within narrow time windows that increase detection likelihood.

Network Segmentation and Microsegmentation: Five organizations had implemented network segmentation isolating Azure AD Connect servers in dedicated VLANs with firewall rules restricting lateral movement. Three of these deployments successfully contained breaches: attackers achieved initial compromise and even domain controller access, but could not reach the isolated sync infrastructure, limiting cloud escalation. The two failures occurred in cases where segmentation was implemented but firewall rules permitted 'administrative' traffic from management subnets, subnets that attackers had also compromised.

This finding underscores a critical point: network segmentation must be absolute for Tier 0 assets. Permitting any lateral movement pathway, even from presumed 'administrative' zones, undermines the control's effectiveness. The most successful implementation observed was a zero-trust microsegmentation

approach that required authenticated, logged connections even from within the administrative network segment.

Synchronization-Specific Monitoring: Only one organization in the sample had implemented real-time monitoring of Azure AD Connect synchronization logs with alerting on anomalous object creations, attribute modifications, or writeback events (Semperis, 2024). This organization detected an intrusion during the lateral movement phase when attackers attempted to create a malicious sync rule. The alert triggered incident response before cloud compromise occurred.

The extreme rarity of this control (5% deployment) is striking given its demonstrated effectiveness. This suggests a significant awareness gap: organizations monitor cloud authentication events and on-premises domain controller logs, but fail to recognize that the synchronization pipeline itself requires dedicated monitoring. The breach cases show that attackers understand this blind spot and deliberately operate within the sync infrastructure to avoid triggering standard security alerts.

4.4.3 Synthesis: The Efficacy-Deployment Gap

The mitigation strategy analysis reveals a troubling disconnect between what organizations deploy and what actually prevents breaches in hybrid AD environments. Standard controls, MFA, conditional access policies, achieve high deployment rates (68-79%) but zero effectiveness against sophisticated attackers who understand hybrid architectures. Meanwhile, controls specifically designed to protect hybrid infrastructure, PAM, network segmentation, sync monitoring, demonstrate 50-100% effectiveness when deployed but are implemented in fewer than 26% of cases.

This efficacy-deployment gap appears driven by three factors:

- **Compliance Orientation:** Organizations prioritize controls that satisfy regulatory requirements and security frameworks (e.g., MFA for NIST 800-53, conditional access for CIS benchmarks) regardless of their effectiveness against actual threat patterns.

- **Operational Friction:** High-efficacy controls like PAM and network segmentation impose operational overhead, requiring approval workflows, changing administrative processes, and potentially impacting system availability during implementation. This friction deters deployment despite security benefits.
- **Awareness Deficit:** The extreme rarity of synchronization-specific monitoring (5%) suggests many organizations do not recognize hybrid architectures as distinct security domains requiring specialized controls. They apply general-purpose security measures (MFA, firewalls, antivirus) without addressing hybrid-specific attack vectors.

These findings comprehensively address RQ3: effective hybrid AD security requires abandoning checkbox compliance in favor of controls specifically designed to protect synchronization infrastructure and credential flows. The controls that work, PAM, segmentation, sync monitoring, share a common characteristic: they disrupt the attack patterns documented in RQ1 and RQ2 by eliminating standing privileges, isolating critical assets, and monitoring the trust boundaries that attackers exploit.

4.5 Cross-Case Synthesis: A Taxonomy of Hybrid AD Attack Propagation

Integrating findings from all three research questions reveals a coherent attack progression model that characterizes hybrid AD compromises. This model, derived from patterns observed across 19 incidents, consists of five phases:

Phase 1: Initial Access – Establishing a Foothold

Attackers gain initial access through diverse vectors: supply chain compromise (SolarWinds), social engineering (MGM, Caesars, Scattered Spider), vulnerability exploitation (Kaseya, VMware ESXi), or credential theft (LastPass, NVIDIA). Notably, the initial access vector showed no correlation with subsequent attack sophistication, unsophisticated phishing campaigns led to equally devastating outcomes as nation-state supply chain compromises once hybrid infrastructure was reached. (Mandiant, 2021; Krebs, 2023; CISA, 2021)

Phase 2: Privilege Escalation – Targeting Hybrid Infrastructure

After initial access, attackers systematically escalate toward one of three hybrid privilege pivots: Azure AD Connect servers (14 incidents), PTA agents (2 incidents), or ADFS servers (3 incidents). This targeting is deliberate and rapid, median time from initial access to hybrid infrastructure compromise was 8 days. Attackers use standard reconnaissance tools (BloodHound, ADRecon, PowerView) to identify synchronization servers, then exploit weak access controls (no network segmentation, standing administrative privileges) to compromise them. (Sygnia, 2024; Diericx, 2024)

Phase 3: Credential Extraction and Reuse

With hybrid infrastructure access, attackers extract credentials usable across architectural boundaries: NTLM hashes from Azure AD Connect memory, PTA agent manipulation capabilities, or SAML token-signing certificates from ADFS servers. These credentials are immediately reused, often within hours, to authenticate to both cloud services (Microsoft 365, Azure portals) and on-premises systems (domain controllers, file servers). This dual-environment access is the defining characteristic of hybrid attacks. (Cymulate, 2025; Semperis, 2024)

Phase 4: Bidirectional Escalation

Attackers leverage their hybrid position to escalate in both directions simultaneously. Cloud-to-on-premises: creating privileged groups in Entra ID and using object writeback to grant Domain Admin rights. On-premises-to-cloud: injecting malicious objects into synchronization pipelines to establish persistent cloud access. Three incidents demonstrated full bidirectional control where attackers could escalate privileges in either environment by manipulating the other, essentially using hybrid architecture as a privilege escalation platform. (Diericx, 2024; Mandiant, 2024)

Phase 5: Objective Execution and Persistence

With complete environment control, attackers execute their objectives: ransomware deployment (9 incidents), data exfiltration (7 incidents), or long-term espionage (3 incidents). Critically, they establish persistence mechanisms that survive standard incident response: forged SAML tokens valid for months, synchronized backdoor accounts that appear legitimate, and modified synchronization rules that recreate access even after password resets. This persistence explains why median dwell time (initial access to detection) was 47 days, attackers embedded themselves in the trusted synchronization infrastructure rather than maintaining noisy external persistence mechanisms.

Phase	Attacker Actions	Critical Enablers	Detection Difficulty	Median Duration
1. Initial Access	Supply chain, phishing, vuln exploit	Weak perimeter, unpatched systems	Moderate	Day 0 (entry point)
2. Priv Escalation	Recon, lateral movement to sync infra	No segmentation, standing admin privs	Difficult	Days 1-8
3. Cred Extract	Hash dump, PTA manipulation, token theft	Sync server compromise, no monitoring	Very difficult	Days 8-10
4. Bidirectional Esc	Writeback abuse, object injection	Unconstrained sync trust, CAP gaps	Extremely difficult	Days 10-15
5. Obj Execution	Ransomware, exfiltration, espionage	Complete env control, persistence	Variable (often too late)	Days 15-47

Table 6: Five-Phase Hybrid AD Attack Progression Model

This taxonomy synthesizes the empirical patterns observed across all breach cases, providing a predictive model for hybrid AD attack progression. The model's utility lies not in predicting when attacks will occur, but in identifying which phase an ongoing intrusion has reached and which controls can still

disrupt it. For example, robust network segmentation remains effective through Phase 2, but once Phase 3 begins (credential extraction), segmentation provides minimal protection, requiring a shift to monitoring and response strategies.

Chapter 5: Discussion

5.1 Introduction

This chapter interprets the findings presented in Chapter 4, contextualizing them within the existing literature and examining their implications for both academic understanding and practical security operations. The discussion is structured around three core areas: interpretation of results in relation to the research questions, comparison with established literature, and critical examination of the study's contributions and limitations.

The empirical analysis of 19 hybrid Active Directory breaches revealed patterns that both confirm and challenge existing assumptions about identity security. Most significantly, the findings expose a critical disconnect between widely deployed security controls and actual attacker behavior, what this study terms the "efficacy-deployment gap." This chapter explores why this gap exists, what it reveals about organizational security priorities, and how emerging approaches like Identity Threat Detection and Response (ITDR) may address systemic vulnerabilities that traditional controls fail to mitigate.

5.2 Interpretation of Results

5.2.1 Synchronization as the Dominant Attack Vector (RQ1)

The finding that 74% of incidents explicitly exploited synchronization mechanisms validates the core hypothesis that hybrid architectures create privileged attack conduits between on-premises and cloud environments. However, the research reveals a more nuanced reality than anticipated: synchronization

infrastructure is not merely *a* target, it is *the* target for sophisticated adversaries who understand hybrid environments.

Three specific patterns warrant interpretation:

Azure AD Connect as Tier 0 Asset, In Theory Only. Despite possessing Enterprise Admin and Global Administrator privileges simultaneously, Azure AD Connect servers were treated as standard infrastructure in every analyzed incident where deployment details were disclosed. No case documented dedicated VLANs, just-in-time access requirements, or real-time monitoring, controls universally applied to domain controllers. This treatment disparity reveals a fundamental misunderstanding: organizations recognize domain controllers as Tier 0 assets because they control on-premises authentication, but fail to recognize that Azure AD Connect servers control authentication *across both architectures*. This cognitive gap, treating a bridge as a single-environment component, enables the attack patterns documented in this study.

Bidirectional Escalation Inverts Traditional Attack Flows. The emergence of cloud-to-on-premises attacks via object writeback (21% of incidents) represents a paradigm shift. Traditional AD security models assume on-premises compromise leads to cloud exposure through credential synchronization. However, cases like MGM Resorts and Scattered Spider demonstrate that attackers increasingly achieve initial access through cloud identity portals (often via social engineering) and then pivot *downward* to on-premises infrastructure. This inversion matters because defensive investments remain skewed toward protecting on-premises environments from external threats, leaving cloud-to-on-premises pathways under-monitored and poorly controlled. The bidirectional nature of hybrid attacks renders unidirectional defensive models obsolete.

Synchronization Trust is Assumed, Not Validated. The consistent success of malicious object injection attacks (26% of incidents) reveals that synchronized identities bypass security controls applied to native cloud identities. Conditional access policies, anomaly detection systems, and risk-based

authentication, all designed to scrutinize cloud authentication, frequently exempt synchronized accounts under the assumption that on-premises AD validation is sufficient. Attackers exploit this exemption by compromising synchronization pipelines rather than individual accounts. This finding suggests organizations have architecturally embedded a trust assumption that attackers systematically violate.

5.2.2 Credential Universality as Architectural Vulnerability (RQ2)

The credential abuse patterns documented in Chapter 4 reveal that hybrid AD implementations treat identities as "universal tokens", credentials or authentication artifacts valid in one environment grant access across architectural boundaries with minimal additional verification. This universality manifests in three specific failure modes:

1. **NTLM Hash Persistence Across Architectures.** The finding that NTLM hashes extracted from Azure AD Connect servers remain valid for both cloud API authentication and on-premises lateral movement until password change contradicts the theoretical security benefit of password hash synchronization. Microsoft's guidance emphasizes that PHS stores "salted and hashed" credentials that cannot be reversed, technically accurate but operationally misleading. Attackers do not need to reverse hashes; they reuse them directly through pass-the-hash techniques for on-premises movement and through Azure AD's NTLM-over-HTTPS authentication for cloud access. The 45-60 day average delay in password rotation observed across incidents extends this vulnerability window substantially. (Microsoft, 2024)
2. **PTA Agent Manipulation as Authentication Bypass.** The "Double Agent" vulnerability identified by Cymulate and observed in 11% of analyzed incidents represents an architectural rather than implementation flaw. Pass-through Authentication was designed to address PHS security concerns by keeping password hashes on-premises, a genuine security improvement for credential storage. However, by making the PTA agent a trusted authentication proxy, the architecture shifts the attack surface from credential theft to *trust relationship manipulation*. Once

compromised, a PTA agent becomes a bypass mechanism for the entire authentication stack, including MFA, conditional access, and risk-based controls.

3. **MFA Enforcement Gaps at Trust Boundaries.** The 100% failure rate of MFA in preventing breaches, despite 79% deployment, requires careful interpretation to avoid the incorrect conclusion that MFA is ineffective. The data shows MFA *when consistently enforced* significantly raises attacker costs (evidenced by sophisticated social engineering campaigns like Scattered Spider specifically developed to bypass MFA).

However, enforcement gaps at hybrid trust boundaries systematically undermine MFA's value:

- **Legacy protocol exemptions:** 42% of incidents involved authentication via SMTP, POP3, or IMAP protocols that do not support MFA, with attackers specifically targeting these legacy pathways. (Microsoft Security Response Center, 2024; LastPass, 2023)
- **Synchronized account exemptions:** 16% of incidents documented conditional access policies that exempted synchronized accounts from MFA requirements. (Secureworks, 2023)
- **Service principal and application identity gaps:** 21% of incidents involved authentication using service principals or application identities that lacked MFA enforcement.

These patterns reveal that MFA functions as a control point, not a control surface. It protects individual authentication events but does not secure the trust relationships and credential flows that enable cross-architecture movement.

5.2.3 The Efficacy-Deployment Gap and Control Selection Failures (RQ3)

The inverse correlation between control deployment rates and effectiveness represents the study's most counterintuitive finding: the most widely deployed controls (MFA 79%, CAPs 68%) demonstrated zero effectiveness in preventing sophisticated hybrid attacks, while rarely deployed controls (PAM 16%,

network segmentation 26%, synchronization monitoring 5%) showed 50-100% effectiveness when properly implemented.

This efficacy-deployment gap cannot be explained by technical complexity alone. Instead, three organizational factors appear to drive this paradoxical pattern:

Compliance Checkbox Mentality. MFA and conditional access policies satisfy explicit requirements in security frameworks (NIST 800-53, CIS Controls, ISO 27001) and regulatory mandates. Organizations deploy these controls to achieve compliance certification, not necessarily to address actual threat patterns. Critically, compliance frameworks rarely specify *how* MFA must be implemented, enabling deployments that satisfy auditors but fail against adversaries. (NIST, 2024; CIS, 2024)

Operational Friction Avoidance. PAM, network segmentation, and synchronization monitoring impose operational costs: approval workflows slow administrative tasks, network segmentation complicates troubleshooting, and sync monitoring generates alerts requiring investigation. These friction points create organizational resistance during implementation. MFA and conditional access policies, once configured, operate transparently without ongoing operational overhead. Organizations demonstrably prefer controls that minimize disruption over controls that maximize security.

Hybrid Architecture Blind Spot. The 5% deployment rate for synchronization-specific monitoring despite its 100% effectiveness in the single documented cases suggests many organizations do not conceptualize hybrid environments as distinct security domains requiring specialized controls. They apply on-premises AD controls and cloud controls without addressing the *interstitial space* where synchronization occurs. This blind spot persists in practitioner awareness, not just technology availability. (Chen et al., 2024; CounterCraft Security, 2025).

5.2.4 Identity Threat Detection and Response (ITDR) as Systemic Mitigation

The findings validate the emerging ITDR approach continuous monitoring and automated response to identity-based threats as particularly well-suited to hybrid environments where preventive controls demonstrate systematic failure. ITDR addresses the three core vulnerability patterns identified in this study:

Monitoring Synchronization Trust Violations. ITDR platforms specifically designed for hybrid environments (e.g., Semperis Directory Services Protector, Purple Knight) instrument the synchronization pipeline itself, generating alerts on: malicious object injection attempts, synchronization rule modifications, writeback operations that elevate privileges, and anomalous authentication patterns involving synchronized accounts. The single case in the dataset where synchronization monitoring detected an attack during lateral movement phase validates this approach: the organization's ITDR solution flagged a new sync rule creation as anomalous, triggering incident response *before* cloud compromise occurred. (Semperis, 2024; Gartner, 2024)

Behavioral Analytics for Credential Abuse Detection. ITDR's behavioral analysis capabilities address credential universality risks by establishing baselines for how credentials are used across architectures and alerting on deviations: synchronized accounts authenticating to unusual cloud services, NTLM authentication originating from geographic locations inconsistent with user profiles, PTA agents processing authentication requests at abnormal volumes, and privilege escalations that correlate with recent synchronization events. (Semperis, 2024).

Automated Containment Responses. ITDR's automated response capabilities (automatically disabling compromised accounts, quarantining suspicious sync rules, isolating affected Azure AD Connect servers) address the speed differential between attacker movement and defender response. The median 8-day window between initial access and hybrid infrastructure compromise observed in the dataset suggests manual incident response processes are too slow to prevent escalation.

However, ITDR's limited deployment (documented in only 2 incidents) and relative immaturity as a product category prevent definitive conclusions about real-world effectiveness. The study's findings support ITDR's theoretical promise but cannot empirically validate it at scale.

5.3 Comparison with Existing Literature

5.3.1 Validation of Hybrid-Specific Vulnerability Research

The findings strongly corroborate Diericx's (2024) research demonstrating 100% Azure AD compromise success rates via compromised Azure AD Connect servers. This study extends Diericx's lab-based analysis by documenting the same attack pattern in 14 real-world incidents, confirming that the theoretical vulnerability translates to operational exploitation. Critically, the study identifies *why* this vulnerability persists: not because organizations lack knowledge of the risk, but because they fail to implement compensating controls (Tier 0 segmentation, PAM, monitoring) that would mitigate it.

Similarly, Sygnia's (2024) technical analysis of Azure AD Connect attack vectors is comprehensively validated. However, the case study data reveals that sophisticated memory extraction techniques are less commonly employed than anticipated. Instead, attackers more frequently exploit weak access controls to the Azure AD Connect server and then use built-in administrative tools to extract credentials. This finding suggests defensive priorities should emphasize *access control* to synchronization infrastructure over *technical hardening* of the synchronization process itself.

5.3.2 Divergence from Traditional AD Security Models

The findings challenge assumptions embedded in traditional AD security literature, which predominantly focuses on on-premises attack patterns. Kumar et al.'s (2024) comprehensive AD security testing framework emphasizes domain controller hardening, Kerberos security, and privilege escalation prevention all valid for isolated on-premises environments but insufficient for hybrid architectures.

The study reveals three specific gaps in traditional AD security models:

Perimeter-Centric Defenses Fail. Traditional AD security emphasizes network perimeter controls to limit external access to AD infrastructure. However, 32% of analyzed incidents achieved initial access through legitimate cloud identity portals, bypassing on-premises perimeters entirely. This validates CounterCraft Security's (2025) observation that "cloud identities have become the new perimeter" but extends it: the cloud is not merely an *alternative* perimeter but a *superior* entry point for adversaries targeting hybrid environments.

Golden Ticket Attacks Gain Cloud Amplification. Melnyk and Galchynsky's (2023) research on Golden Ticket detection focuses on on-premises Kerberos traffic analysis. However, 16% of analyzed incidents used Golden Tickets not for on-premises lateral movement but to authenticate to Azure AD Connect servers and manipulate synchronization, thereby achieving *cloud* compromise. This hybrid attack variant evades traditional detection systems that monitor only on-premises domain controller traffic.

Legacy Protocol Vulnerabilities Persist Despite Deprecation Guidance. The Australian Cyber Security Centre (2024) and NIST (2024) recommend disabling NTLMv1, WDigest, and other legacy protocols. However, 42% of analyzed incidents involved legacy protocol exploitation not because organizations were unaware but because hybrid architectures *require* legacy protocol support for backward compatibility. This reveals a tension between security guidance (disable legacy protocols) and architectural reality (hybrid integration often depends on them).

5.3.3 Social Engineering in Hybrid Contexts

The study's findings strongly validate Roberts et al.'s (2023) research identifying hybrid environments as particularly susceptible to human-operated attacks. The Scattered Spider campaigns (affecting 6 of 19 analyzed incidents) exemplify this pattern: attackers used vishing, MFA fatigue, and helpdesk impersonation not to compromise on-premises AD directly, but to gain cloud administrator access, which they subsequently leveraged for on-premises infiltration.

However, the case study data reveals nuance not captured in Roberts et al.'s analysis: social engineering attacks targeting hybrid environments are *qualitatively different* from traditional phishing campaigns. Rather than seeking credentials for lateral movement within a single environment, attackers specifically target *architectural bridge roles* IT support staff who manage Azure AD Connect, helpdesk personnel who can reset MFA registrations, and administrators who hold privileged roles in both directories. This targeting precision suggests adversaries perform reconnaissance to map hybrid architecture and identify high-value social engineering targets based on their position in the identity synchronization process.

5.3.4 Zero Trust Implementation Gaps

The findings provide empirical validation for Lepide's (2024) and BeyondTrust's (2025) arguments that Zero Trust Architecture is particularly critical for hybrid environments. However, the study reveals that Zero Trust adoption is not binary but spectrum-based, with most organizations implementing Zero Trust *partially*:

- 68% deployed conditional access policies (a Zero Trust component) but applied them inconsistently across identity types
- Only 16% implemented just-in-time privileged access (core Zero Trust principle)
- 5% deployed continuous monitoring of synchronization trust relationships (Zero Trust's "never trust, always verify")

Critically, the finding that network segmentation and PAM both Zero Trust architectural components demonstrated high effectiveness when deployed suggests that Zero Trust principles *do* work against hybrid attacks, but only when applied *holistically* across all architectural components, including the synchronization layer.

5.4 Theoretical and Practical Implications

5.4.1 Theoretical Contributions

Cross-Architecture Threat Taxonomy. This study develops the first empirically grounded taxonomy of hybrid AD attack patterns, categorizing attacks by direction (on-premises→cloud, cloud→on-premises, bidirectional), vector (synchronization exploitation, credential abuse, social engineering), and phase (initial access, privilege escalation, credential extraction, bidirectional escalation, objective execution). This taxonomy enables more precise threat modeling and provides a structured framework for future hybrid security research.

Efficacy-Deployment Gap as Research Construct. The observed inverse correlation between security control deployment and effectiveness introduces a construct with explanatory power beyond hybrid AD security. The gap may characterize security domains where compliance drivers, operational friction, and emerging threats create selection pressures favoring easily deployable but less effective controls. This construct warrants investigation in other security domains (cloud security, IoT security, OT security).

Synchronization Trust as Fundamental Design Flaw. The study provides empirical evidence that treating synchronization pipelines as "trusted" architectural components rather than attack surfaces constitutes a fundamental design flaw in hybrid identity systems. This finding challenges the implicit trust models embedded in current hybrid AD architectures and suggests that future identity federation designs should instrument and monitor trust relationships rather than assuming their integrity.

5.4.2 Practical Implications for Security Practitioners

Reclassification of Azure AD Connect as Tier 0 Asset. Organizations must immediately reclassify Azure AD Connect servers (and by extension, PTA agents and ADFS servers) as Tier 0 assets requiring equivalent protections to domain controllers: dedicated administrative workstations, just-in-time

access controls, network microsegmentation, and continuous monitoring. This reclassification has budgetary and operational implications but is non-negotiable given the 74% exploitation rate documented in this study.

Prioritization of Hybrid-Specific Controls Over Generic Best Practices. Security teams should reassess control investment priorities, deprioritizing incremental improvements to widely deployed but ineffective controls (e.g., expanding MFA to additional user populations) in favor of deploying hybrid-specific controls with demonstrated effectiveness (synchronization monitoring, PAM, network segmentation). This recommendation challenges conventional wisdom that emphasizes control breadth over control depth.

Implementation of ITDR Capabilities. Organizations operating hybrid AD should evaluate Identity Threat Detection and Response platforms that specifically address synchronization pipeline monitoring, cross-architecture credential abuse detection, and automated containment. While ITDR remains an emerging product category, the study's findings suggest it directly addresses the attack patterns that traditional SIEM and EDR solutions fail to detect. Early adoption by security-mature organizations is warranted given the 47-day median dwell time observed in incidents lacking ITDR capabilities.

Architectural Security Requirements for Hybrid Deployments. Organizations planning new hybrid AD deployments should incorporate security requirements during architecture design: synchronization servers must reside in isolated network segments, privileged accounts used for synchronization must be dedicated, synchronization activities must be logged and monitored with automated alerting, and object writeback must be disabled unless specifically required and then tightly scoped to non-privileged objects.

5.5 Limitations of the Study

5.5.1 Methodological Limitations

Reliance on Secondary Data. The study's exclusive use of publicly disclosed breach data introduces multiple limitations. Reporting bias likely overrepresents high-profile incidents affecting large organizations while underrepresenting breaches affecting smaller organizations or those that remain confidential. The sample may skew toward attacks that were *detected* and *disclosed*, systematically excluding stealthy long-term compromises that evaded detection. Additionally, the level of technical detail varies significantly across sources, constraining comparative analysis depth.

No Experimental Validation. The findings derive from observational analysis of real-world incidents, not controlled experiments. While this approach provides ecological validity (attacks actually occurred in production environments), it precludes causal claims about control effectiveness. The correlation between PAM deployment and attack containment, for example, suggests but does not *prove* that PAM was the causal factor organizations deploying PAM may differ in other security practices that contributed to successful defense.

Temporal and Technical Scope Boundaries. The 2017-2025 timeframe ensures findings reflect modern hybrid AD configurations but excludes earlier incidents and future threat evolution. Microsoft's ongoing architectural changes particularly the shift toward cloud-native identity with Entra ID features superseding traditional AD capabilities, may reduce the relevance of synchronization-based attacks over time. Additionally, the focus on Microsoft hybrid AD excludes third-party identity providers, non-Windows endpoints, and alternative directory services, limiting generalizability to heterogeneous identity environments.

Qualitative Methodology Constraints. As a qualitative case study, the research prioritizes depth and contextual understanding over statistical generalizability. The sample of 19 incidents enables pattern identification and analytical generalization to theoretical propositions, but cannot support claims about

attack prevalence, control effectiveness rates, or population-level statistics. Percentages reported (e.g., "74% of incidents exploited synchronization") describe the study sample, not the universe of all hybrid breaches.

5.5.2 Potential Biases and Confounding Factors

Sophistication Bias. Publicly disclosed incidents disproportionately involve sophisticated threat actors (nation-states, ransomware operations) whose tactics may not represent typical attacker behavior. Less sophisticated opportunistic attacks likely occur more frequently but garner less attention, potentially skewing the findings toward advanced persistent threat patterns.

Organizational Context Confounding. The effectiveness findings (PAM, segmentation, monitoring) may reflect organizational security maturity rather than control-specific efficacy. Organizations that deploy specialized hybrid controls likely differ systematically in security budget, staffing expertise, executive support, and overall security culture from organizations that rely solely on standard controls. These confounding factors may contribute to observed effectiveness differences beyond the controls themselves.

Vendor Reporting Perspective. Many analyzed incidents were documented by cybersecurity vendors who conducted incident response. These vendors may emphasize technical attack details that showcase their detection capabilities while downplaying organizational or process failures that contributed to compromise. Additionally, vendor reports may describe "what happened" with greater certainty than forensic evidence actually supports.

5.5.3 Generalizability Constraints

The findings are most directly applicable to:

- Enterprise organizations (500+ employees) operating Microsoft hybrid AD environments
- Azure AD Connect, Pass-through Authentication, or ADFS hybrid configurations

- Windows Server 2016 or later on-premises AD implementations
- Organizations in mature cybersecurity threat landscapes (primarily North America, Europe)

Generalizability is uncertain for:

- Small and medium businesses with different security resource constraints
- Hybrid identity environments using third-party integration tools
- Organizations in regions with different threat actor profiles
- Pure cloud-native identity implementations without on-premises AD synchronization

5.6 Recommendations for Future Research

5.6.1 Priority Research Directions

Quantitative Prevalence Studies. Future research should establish base rate statistics for hybrid AD attacks: what percentage of organizations experience synchronization-based compromises annually? What is the probability distribution of dwell times? How do attack success rates vary by hybrid configuration type? Large-scale quantitative studies, potentially through anonymized telemetry from ITDR vendors or security service providers, would complement this study's qualitative findings and enable risk quantification for organizational decision-making.

Experimental Control Effectiveness Testing. Controlled red team experiments comparing mitigation strategies would strengthen causal claims about control effectiveness. Research designs could deploy identical hybrid AD environments with varying security controls, execute standardized attack chains, and measure time-to-compromise, detection rates, and containment effectiveness. Such experiments would isolate control effects from organizational confounding factors present in real-world incidents.

Longitudinal Threat Evolution Analysis. As Microsoft continues evolving Entra ID capabilities and organizations migrate toward cloud-native identity, research tracking how attack patterns adapt is

critical. Will synchronization-based attacks decline as organizations adopt cloud-only identity? Or will attackers shift to new integration points? Longitudinal research would inform long-term architectural security investments.

SMB-Focused Hybrid Security Research. This study's focus on enterprise incidents leaves small and medium business hybrid security under-researched. SMBs face different constraints (limited security staffing, budget pressures, reliance on managed service providers) that may affect both attack patterns and feasible mitigations. Research investigating whether enterprise-derived recommendations translate to SMB contexts would address a significant practitioner need.

5.6.2 Methodological Extensions

Mixed-Methods Hybrid Security Studies. Future research combining qualitative case analysis with quantitative surveys of security practitioners could address this study's limitations. Surveys could establish prevalence of security practices, perceived effectiveness, implementation challenges, and organizational characteristics correlated with control adoption. Integrating quantitative prevalence data with qualitative incident analysis would provide more comprehensive understanding.

Grounded Theory Development for Efficacy-Deployment Gap. The observed inverse correlation between deployment and effectiveness warrants theoretical investigation. Grounded theory methodology, involving practitioner interviews, organizational ethnography, and iterative theory building, could develop explanatory frameworks for why organizations consistently prioritize less effective controls. Such research would have implications beyond hybrid AD security for cybersecurity decision-making broadly.

Comparative Studies Across Identity Platforms. Research comparing security incident patterns across Microsoft hybrid AD, Okta Workforce Identity, Google Cloud Identity, and other enterprise identity platforms would establish whether findings are Microsoft-specific or generalizable to hybrid

identity architectures broadly. Cross-platform comparison would identify universal hybrid security challenges versus vendor-specific vulnerabilities.

5.6.3 Emerging Technology Implications

Zero Trust Architecture Maturity Models for Hybrid Environments. Research developing assessment frameworks for Zero Trust maturity specifically in hybrid identity contexts would operationalize this study's findings. Such frameworks should measure not just control deployment but control effectiveness against documented hybrid attack patterns, enabling organizations to evaluate their security posture objectively.

ITDR Platform Effectiveness Evaluation. As ITDR matures from emerging technology to mainstream product category, comparative research evaluating different vendors' detection capabilities, false positive rates, and containment effectiveness would guide organizational tool selection. Independent academic evaluation of ITDR platforms, analogous to endpoint protection platform testing, would accelerate evidence-based adoption.

AI/ML for Hybrid Threat Detection. Research investigating whether machine learning models can detect synchronization-based attacks from authentication logs, directory modification events, or network traffic would assess feasibility of behavioral detection approaches. Training datasets derived from real-world hybrid breach data could enable development of detection algorithms deployable in security tools.

5.7 Chapter Summary

This chapter interpreted the empirical findings from Chapter 4, contextualized them within existing literature, and examined their theoretical and practical implications. The discussion validates the central proposition that hybrid Active Directory environments create systemically vulnerable attack

surfaces where synchronization mechanisms serve as privileged bridges that adversaries systematically exploit.

Three critical insights emerged. First, the efficacy-deployment gap, where rarely deployed controls demonstrate high effectiveness while widely deployed controls show zero effectiveness, reveals organizational security decision-making prioritizes compliance, operational convenience, and generic best practices over evidence-based mitigation of actual attack patterns. Second, synchronization infrastructure's treatment as "trusted" architectural components rather than attack surfaces constitutes a fundamental design flaw that attackers consistently exploit. Third, the emergence of bidirectional attacks (cloud→on-premises) inverts traditional threat models and renders perimeter-centric defenses obsolete.

The findings strongly support adopting ITDR approaches that instrument synchronization pipelines, detect credential abuse through behavioral analytics, and automate containment responses. While ITDR's limited deployment in the dataset prevents definitive effectiveness claims, the single documented case of successful attack detection through synchronization monitoring validates its theoretical promise.

The study's limitations, reliance on secondary data, lack of experimental validation, qualitative methodology constraints, are acknowledged transparently. These limitations do not undermine the validity of pattern identification or the study's contribution to hybrid AD security knowledge, but they constrain generalizability and causal claims. Future research should address these gaps through quantitative prevalence studies, experimental control testing, and longitudinal threat evolution analysis.

Chapter 6 will synthesize the dissertation's contributions, present evidence-based recommendations for hardening hybrid AD infrastructure, and conclude with reflections on the research process and implications for the evolving field of identity security

Chapter 6: Conclusion

6.1 Summary of Key Findings

This qualitative multiple-case study analyzed 19 documented hybrid Active Directory breaches (2017–2025) to investigate cross-architecture attack propagation. The research establishes that hybrid AD environments create systemically vulnerable attack surfaces where synchronization infrastructure serves as privileged attack conduits.

Three critical findings emerged:

- **Synchronization mechanisms as attack vectors (RQ1):** In 74% of incidents, attackers explicitly targeted Azure AD Connect servers, synchronization rules, or object writeback features to achieve bidirectional escalation between cloud and on-premises environments. Azure AD Connect servers, granted Enterprise Admin and Global Administrator privileges, represent catastrophic single points of failure.
- **Credential universality enabling cross-architecture abuse (RQ2):** NTLM hashes, PTA agents, and SAML tokens function as universal authentication tokens across architectural boundaries. Attackers exploit credential reuse through PHS memory extraction, PTA agent manipulation, and SAML certificate theft. Inconsistent MFA enforcement creates exploitable gaps where sophisticated attackers bypass authentication controls.
- **The efficacy-deployment gap (RQ3):** Widely deployed controls (MFA 79%, Conditional Access 68%) demonstrated zero effectiveness against sophisticated attackers, while rarely deployed controls (PAM 16%, network segmentation 26%, synchronization monitoring 5%) showed 50–100% effectiveness when properly implemented. This inverse correlation reveals that organizations prioritize compliance-driven controls over evidence-based mitigations addressing actual attack patterns.

The synthesized five-phase attack progression model (initial access → privilege escalation → credential extraction → bidirectional escalation → objective execution) demonstrates that hybrid compromises follow predictable patterns, with *median dwell time of 47 days* before detection.

6.2 Contributions to Knowledge

Theoretical contributions. This research advances identity security scholarship by developing an empirically grounded cross-architecture threat model. The study demonstrates that synchronization infrastructure constitutes a distinct security domain requiring specialized analytical frameworks, rejecting the assumption that on-premises and cloud controls independently secure hybrid environments. The trust boundary exploitation theory, validated through 19 real-world incidents, establishes that attackers systematically target the interstitial space where implicit trust enables privileged movement. The hybrid-specific attack propagation taxonomy provides a predictive framework for security practitioners and researchers.

Empirical contributions. The dataset of 19 verified hybrid AD breaches represents the largest qualitative analysis of cross-architecture attacks in published research. By synthesizing incident data from nation-state campaigns (SolarWinds, Microsoft corporate breach), ransomware operations (BlackCat, ALPHV), and financially motivated attacks (Scattered Spider), the study demonstrates that hybrid vulnerabilities transcend attacker sophistication, industry sector, and organizational scale. The analysis challenges prevailing assumptions by documenting bidirectional attack patterns (cloud-to-on-premises escalation), contradicting traditional perimeter-based threat models.

Methodological contributions. The conceptual framework integrating Zero Trust Architecture, MITRE ATT&CK, and trust boundary exploitation theory demonstrates methodological rigor in qualitative security research. The five-phase thematic analysis with explicit source triangulation, negative case analysis, and audit trails establishes reproducible analytical standards for breach case studies, addressing methodological weaknesses in existing incident-based cybersecurity research.

6.3 Practical Implications

Immediate reclassification of synchronization infrastructure. Organizations must reclassify Azure AD Connect servers, PTA agents, and ADFS servers as Tier 0 assets requiring protections equivalent to domain controllers: dedicated administrative workstations with no internet access, just-in-time privileged access management, network microsegmentation isolating synchronization traffic, and continuous behavioral monitoring. This reclassification has budgetary implications but is non-negotiable given empirically documented exploitation rates.

Prioritization of hybrid-specific controls. Security teams should deprioritize incremental improvements to ineffective controls and instead deploy hybrid-specific mitigations: implement synchronization monitoring (ITDR platforms instrumenting Azure AD Connect pipelines), enforce network segmentation preventing lateral movement to sync infrastructure, deploy comprehensive PAM eliminating standing administrative privileges, and harden object writeback configurations restricting bidirectional privilege escalation.

Adoption of Identity Threat Detection and Response. Organizations should evaluate ITDR platforms providing synchronization pipeline monitoring, cross-architecture credential abuse detection through behavioral analytics, and automated containment responses. While ITDR remains an emerging product category, the empirically documented 47-day median dwell time demonstrates that manual incident response processes are inadequate for hybrid threats. Early adoption by security-mature organizations is warranted. (NIST, 2024; Gartner, 2024)

Architectural security requirements. Organizations planning hybrid AD deployments must incorporate security-by-design principles: synchronization servers in isolated VLANs with restrictive firewall rules, dedicated service accounts with no interactive logon rights, comprehensive logging to immutable storage, object writeback disabled unless operationally critical (and then scoped to non-

privileged objects), and MFA enforcement for all cloud administrative accounts without exception policies.

6.4 Final Thoughts

Hybrid Active Directory environments represent the dominant enterprise identity architecture, yet security frameworks systematically underestimate their vulnerability. This research demonstrates that the convergence of on-premises and cloud identity domains creates exploitable trust boundaries that attackers predictably target. The efficacy-deployment gap, where organizations invest in compliance-driven controls rather than evidence-based mitigations, perpetuates systemic weaknesses despite widespread awareness of hybrid AD threats.

The findings strongly support Zero Trust Architecture principles that explicitly reject architectural trust boundaries. Organizations cannot secure hybrid environments by independently hardening on-premises and cloud components, they must instrument, monitor, and actively defend the synchronization infrastructure that bridges these domains. Identity Threat Detection and Response represents a paradigm shift from preventive controls toward continuous monitoring and automated containment, directly addressing the behavioral patterns documented in this research.

This dissertation contributes empirically grounded insights to an underresearched security domain of critical importance to global enterprises. While constrained by qualitative methodology and reliance on secondary data, the research establishes foundational understanding of cross-architecture attack propagation that future studies can build upon through quantitative prevalence analysis, experimental control validation, and longitudinal threat evolution tracking.

As enterprises continue migrating toward cloud-native identity architectures, the hybrid transition period presents maximum vulnerability. Organizations operating hybrid AD must recognize that synchronization infrastructure represents their most critical attack surface and allocate security resources accordingly. The evidence is unambiguous: hybrid AD breaches succeed by exploiting trust, and only

explicit distrust, operationalized through monitoring, segmentation, and least privilege, can disrupt these attack patterns.

References

Australian Cyber Security Centre (2024) *Detecting and mitigating Active Directory compromises*.

Canberra: Commonwealth of Australia. Available at: <https://www.cyber.gov.au/sites/default/files/2024-09/PROTECT-Detecting-and-Mitigating-Active-Directory-Compromises.pdf> (Accessed: 10 October 2025).

BeyondTrust (2025) *What is Active Directory security?* Available at:

<https://www.beyondtrust.com/resources/glossary/active-directory-security> (Accessed: 3 September 2025).

Braun, V. and Clarke, V. (2006) 'Using thematic analysis in psychology', *Qualitative Research in Psychology*, 3(2), pp. 77-101. doi: 10.1191/1478088706qp063oa.

Chen, L., Williams, K. and Martinez, A. (2024) 'Hybrid Active Directory security: cross-architecture threat analysis', *Journal of Computer Security*, 31(2), pp. 123-145. doi: 10.3233/JCS-230145.

CounterCraft Security (2025) *Active Directory & Entra ID attacks: the hybrid cloud conundrum*.

Available at: <https://www.countercraftsec.com/blog/active-directory-attacks-hybrid-cloud-conundrum/> (Accessed: 26 August 2025).

CrowdStrike (2025) *Active Directory security report: enterprise identity management threats*.

Sunnyvale, CA: CrowdStrike Inc.

Cymulate (2025) *Double agent: exploiting pass-through authentication credential validation in Azure*

AD. Available at: <https://cymulate.com/blog/exploiting-pta-credential-validation-in-azure-ad/> (Accessed: 22 June 2025).

Diericx, L. (2024) *What are the potential security risks of a hybrid Azure Active Directory setup?* The Security Factory. Available at: <https://thesecurityfactory.be/what-are-the-potential-security-risks-of-a-hybrid-azure-active-directory-setup-in-the-event-of-an-on-premise-active-directory-compromise/> (Accessed: 15 July 2025).

JumpCloud (2025) *What is DCSync? Critical Active Directory attack explained.* Available at: <https://jumpcloud.com/it-index/what-is-dcsync-critical-active-directory-attack-explained> (Accessed: 5 August 2025).

Kumar, S., Patel, R. and Johnson, M. (2024) 'Comprehensive security testing of Active Directory implementations: vulnerability assessment and mitigation strategies', *Journal of Cybersecurity Research*, 15(3), pp. 45-67. doi: 10.1016/j.jcr.2024.045.

Lepide (2024) *How to implement Zero Trust architecture in Active Directory.* Available at: <https://www.lepide.com/blog/how-to-implement-zero-trust-architecture-in-active-directory/> (Accessed: 30 August 2025).

Lincoln, Y.S. and Guba, E.G. (1985) *Naturalistic inquiry.* Beverly Hills, CA: Sage Publications.

Melnyk, O. and Galchynsky, V. (2023) 'Advanced detection methods for Golden Ticket attacks in Active Directory environments', *International Journal of Information Security*, 22(4), pp. 891-908. doi: 10.1007/s10207-023-00691-2.

Microsoft Security Response Center (2024) *Security breach disclosure: nation-state attack on Microsoft corporate systems.* Redmond, WA: Microsoft Corporation. Available at: <https://msrc.microsoft.com/blog/2024/01/microsoft-actions-following-attack-by-nation-state-actor-midnight-blizzard/> (Accessed: 2 July 2025).

National Institute of Standards and Technology (2024) *Zero Trust architecture.* NIST Special Publication 800-207. Gaithersburg, MD: U.S. Department of Commerce. doi: 10.6028/NIST.SP.800-207.

Oasis Security (2024) *Oasis Security Research Team discovers Microsoft Azure MFA bypass*. Available at: <https://www.oasis.security/blog/oasis-security-research-team-discovers-microsoft-azure-mfa-bypass> (Accessed: 6 September 2025).

Palo Alto Networks Unit 42 (2024) *Next-generation Kerberos attacks: Unit 42 threat intelligence report*. Santa Clara, CA: Palo Alto Networks Inc.

Patton, M.Q. (2015) *Qualitative research & evaluation methods*. 4th edn. Thousand Oaks, CA: SAGE Publications.

Roberts, K., Ahmed, N. and Clarke, D. (2023) 'Social engineering in hybrid cloud environments: attack vectors and countermeasures', *Social Engineering Research Quarterly*, 12(4), pp. 89-106. doi: 10.1177/2632666X231198745.

Secureworks (2023) *Azure AD conditional access policy tampering: incident analysis and response*. Atlanta, GA: Dell Secureworks Counter Threat Unit.

Semperis (2023) *Securing hybrid AD environments*. Hoboken, NJ: Semperis Inc. Available at: <https://www.semperis.com/wp-content/uploads/resources-pdfs/resources-securing-hybrid-ad-environments.pdf> (Accessed: 30 June 2025).

Semperis (2024) *Pass-the-hash attack vectors in modern Windows environments*. Hoboken, NJ: Semperis Inc.

Sygnia (2024) *Guarding the bridge: new attack vectors in Azure AD Connect*. Available at: <https://www.sygnia.co/blog/guarding-the-bridge-new-attack-vectors-in-azure-ad-connect/> (Accessed: 19 June 2025).

Syynimaa, N. (2020) 'Exploring Azure Active Directory attack surface with open-source intelligence tools', in *Proceedings of the Nordic Conference on Secure IT Systems*. Helsinki: Springer, pp. 270-285.

Available at: <https://pdfs.semanticscholar.org/270f/1a56a219c67ced83195746fea739c050753e.pdf>

(Accessed: 13 September 2025).

Tanium (2023) *What is Active Directory security? Risks and best practices*. Available at:

<https://www.tanium.com/blog/what-is-active-directory-security/> (Accessed: 17 October 2025).

Yin, R.K. (2018) *Case study research and applications: design and methods*. 6th edn. Thousand Oaks, CA: SAGE Publications.

Specific Breach Incidents Referenced

SolarWinds Breach (2020-2021) Mandiant (2021) *Highly evasive attacker leverages SolarWinds supply chain to compromise multiple global victims with SUNBURST backdoor*. Milpitas, CA:

FireEye/Mandiant. Available at: <https://www.mandiant.com/resources/blog/evasive-attacker-leverages-solarwinds-supply-chain-compromises-with-sunburst-backdoor> (Accessed: 16 June 2025).

Colonial Pipeline (2021) U.S. Department of Justice (2021) *Department of Justice seizes \$2.3 million in cryptocurrency paid to the ransomware extortionists Darkside*. Press Release, 7 June. Available at:

<https://www.justice.gov/opa/pr/departments-justice-seizes-23-million-cryptocurrency-paid-ransomware-extortionists-darkside> (Accessed: 14 July 2025).

Kaseya VSA (2021) CISA (2021) *Kaseya VSA ransomware attack guidance*. Cybersecurity and

Infrastructure Security Agency Alert AA21-195A. Available at: <https://www.cisa.gov/news-events/cybersecurity-advisories/aa21-195a> (Accessed: 17 October 2025).

NotPetya (2017) Greenberg, A. (2018) 'The untold story of NotPetya, the most devastating cyberattack in

history', *Wired*, 22 August. Available at: <https://www.wired.com/story/notpetya-cyberattack-ukraine-russia-code-crashed-the-world/> (Accessed: 8 May 2025).

MGM Resorts & Caesars Entertainment (2023) Krebs, B. (2023) 'MGM Resorts ESXi servers

allegedly encrypted in ransomware attack', *Krebs on Security*, 12 September. Available at:

<https://krebsonsecurity.com/2023/09/mgm-resorts-esxi-servers-allegedly-encrypted-in-ransomware-attack/> (Accessed: 15 October 2025).

Mandiant (2024) 'UNC3944: SMS phishing attacks from an aggressive financially motivated actor', *Mandiant Blog*, 15 January. Available at: <https://www.mandiant.com/resources/blog/unc3944-sms-phishing> (Accessed: 26 August 2025).

Change Healthcare (2024) American Hospital Association (2024) *Change Healthcare cyberattack update*. Chicago, IL: AHA. Available at: <https://www.aha.org/cyber-and-risk/change-healthcare-cyberattack> (Accessed: 24 June 2025).

Scattered Spider Campaign Mandiant (2024) 'Scattered Spider: the modus operandi of the roaming threat actor', *Mandiant Blog*, 22 March. Available at: <https://www.mandiant.com/resources/blog/scattered-spider-modus-operandi> (Accessed: 6 July 2025).

Okta Breaches (2022-2023) Okta (2023) *Recent Okta support case security incident*. San Francisco, CA: Okta Inc. Available at: <https://sec.okta.com/articles/2023/10/recent-okta-support-case-security-incident> (Accessed: 16 September 2025).

LastPass (2022) LastPass (2023) *Notice of recent security incident*. Boston, MA: LastPass. Available at: <https://blog.lastpass.com/2023/03/security-incident-update-recommended-actions/> (Accessed: 21 August 2025).

VMware ESXi Ransomware (2023) CISA (2023) *ESXiArgs ransomware virtual machine recovery guidance*. Cybersecurity and Infrastructure Security Agency Alert AA23-039A. Available at: <https://www.cisa.gov/news-events/cybersecurity-advisories/aa23-039a> (Accessed: 15 September 2025).

Attack Techniques and Frameworks

MITRE (2024) *ATT&CK framework for enterprise*. McLean, VA: The MITRE Corporation. Available at: <https://attack.mitre.org/> (Accessed: 27 July 2025).

Metcalf, S. (2015) 'Attackers can now use Mimikatz to implant skeleton key on domain controllers & backdoor every account', *Active Directory Security Blog*, 27 January. Available at: <https://adsecurity.org/?p=1275> (Accessed: 18 September 2025).

Golden Ticket Attacks Campbell, C. and Adams, J. (2023) 'Golden ticket attacks: detection and mitigation in modern Active Directory environments', *IEEE Security & Privacy*, 21(3), pp. 45-56. doi: 10.1109/MSEC.2023.3271234.

Pass-the-Hash Delpy, B. (2023) *Mimikatz documentation and source code*. Available at: <https://github.com/gentilkiwi/mimikatz> (Accessed: 18 May 2025).

DCSync Attacks Schroeder, W. (2015) 'Mimikatz DCSync usage, exploitation, and detection', *Harmj0y Blog*, 17 September. Available at: <http://www.harmj0y.net/blog/redteaming/mimikatz-and-dcsync-and-extrasids-oh-my/> (Accessed: 7 August 2025).

Golden SAML CyberArk (2017) *Golden SAML: newly discovered attack technique forges authentication to cloud apps*. Newton, MA: CyberArk Labs. Available at: <https://www.cyberark.com/resources/threat-research-blog/golden-saml-newly-discovered-attack-technique-forges-authentication-to-cloud-apps> (Accessed: 19 July 2025).

Diamond Ticket Unit 42 (2024) 'Diamond ticket attacks: forging Kerberos tickets using stolen credentials', *Palo Alto Networks Blog*, 8 March. Available at: <https://unit42.paloaltonetworks.com/diamond-ticket-kerberos/> (Accessed: 18 May 2025).

Australian Government (2024) *Cyber sanction imposed on Russian cybercriminal for 2022 Medibank Private compromise*. Available at: <https://www.cyber.gov.au/about-us/view-all-content/news-and-media/cyber-sanction-imposed-russian-cybercriminal-2022-medibank-private-compromise> (Accessed: 12 June 2025).

CISA (2023) *Review of the attacks associated with Lapsus\$ and related threat groups*. Cyber Safety Review Board. Available at: [https://www.cisa.gov/sites/default/files/2023-08/CSRB_Lapsus\\$_508c.pdf](https://www.cisa.gov/sites/default/files/2023-08/CSRB_Lapsus$_508c.pdf) (Accessed: 10 September 2025).

FBI (2021) *REvil ransomware gang behind JBS cyberattack*. NPR, 3 June. Available at: <https://www.npr.org/2021/06/03/1002819883/revil-a-notorious-ransomware-gang-was-behind-jbs-cyberattack-the-fbi-says> (Accessed: 17 May 2025).

Kojima Industries (2022) *Toyota cyberattack on supplier*. CNN Business, 1 March. Available at: <https://www.cnn.com/2022/03/01/business/toyota-japan-cyberattack-production-restarts-intl-hnk/index.html> (Accessed: 22 July 2025).

Microsoft MSRC (2023) *Microsoft mitigates China-based threat actor Storm-0558 targeting of customer email*. Microsoft Security Response Center, 11 July. Available at: <https://msrc.microsoft.com/blog/2023/07/microsoft-mitigates-china-based-threat-actor-storm-0558-targeting-of-customer-email/> (Accessed: 9 July 2025).

NAO (2017) *Investigation: WannaCry cyber attack and the NHS*. National Audit Office, October. Available at: <https://www.nao.org.uk/wp-content/uploads/2017/10/Investigation-WannaCry-cyber-attack-and-the-NHS.pdf> (Accessed: 4 May 2025).

U.S. State Department (2022) *Reward offers for information to bring Conti ransomware variant co-conspirators to justice*, 6 May. Available at: <https://www.state.gov/> (Accessed: 15 July 2025).

Identity and Access Management

Azure AD Connect Documentation Microsoft (2024) *What is Azure AD Connect?* Microsoft Learn. Available at: <https://learn.microsoft.com/en-us/entra/identity/hybrid/connect/whatis-azure-ad-connect> (Accessed: 21 May 2025).

Pass-through Authentication Microsoft (2024) *Azure Active Directory pass-through authentication*.

Microsoft Learn. Available at: <https://learn.microsoft.com/en-us/entra/identity/hybrid/connect/how-to-connect-pta> (Accessed: 17 June 2025).

Password Hash Synchronization Microsoft (2024) *Implement password hash synchronization with*

Azure AD Connect sync. Microsoft Learn. Available at: <https://learn.microsoft.com/en-us/entra/identity/hybrid/connect/how-to-connect-password-hash-synchronization> (Accessed: 1 May 2025).

Active Directory Federation Services Microsoft (2024) *Active Directory Federation Services overview*.

Microsoft Learn. Available at: <https://learn.microsoft.com/en-us/windows-server/identity/ad-fs/ad-fs-overview> (Accessed: 28 September 2025).

Security Frameworks and Standards

CIS Controls Center for Internet Security (2024) *CIS Controls v8*. East Greenbush, NY: CIS. Available

at: <https://www.cisecurity.org/controls/v8> (Accessed: 21 June 2025).

ISO/IEC 27001 International Organization for Standardization (2022) *ISO/IEC 27001:2022 Information security, cybersecurity and privacy protection*. Geneva: ISO.

PAS 754 - Trustworthy Software British Standards Institution (2024) *PAS 754:2024 Software*

trustworthiness – Governance and management – Guide. London: BSI Standards Limited.

Identity Threat Detection and Response (ITDR)

Gartner (2024) *Market guide for identity threat detection and response*. Stamford, CT: Gartner Inc.

Semperis (2024) *Directory Services Protector: protecting Active Directory and Azure AD*. Hoboken, NJ:

Semperis Inc. Available at: <https://www.semperis.com/directory-services-protector/> (Accessed: 6 August 2025).

Purple Knight (2024) *Active Directory security assessment tool*. Available at: <https://www.purple-knight.com/> (Accessed: 2 June 2025).

Additional Academic Sources

Hybrid Cloud Security Zhang, Y., Liu, W. and Wang, Z. (2023) 'Security challenges in hybrid cloud identity management: a systematic review', *ACM Computing Surveys*, 55(9), pp. 1-38. doi: 10.1145/3571156.

Zero Trust Implementation Rose, S., Borchert, O., Mitchell, S. and Connelly, S. (2020) *Zero Trust architecture*. NIST Special Publication 800-207. Gaithersburg, MD: National Institute of Standards and Technology. doi: 10.6028/NIST.SP.800-207.

Qualitative Research Methods Creswell, J.W. and Poth, C.N. (2018) *Qualitative inquiry and research design: choosing among five approaches*. 4th edn. Thousand Oaks, CA: SAGE Publications.

Case Study Research Stake, R.E. (1995) *The art of case study research*. Thousand Oaks, CA: SAGE Publications.

Thematic Analysis Nowell, L.S., Norris, J.M., White, D.E. and Moules, N.J. (2017) 'Thematic analysis: striving to meet the trustworthiness criteria', *International Journal of Qualitative Methods*, 16(1), pp. 1-13. doi: 10.1177/1609406917733847.

Threat Actor Groups

APT29/Midnight Blizzard Mandiant (2023) *APT29 continues to target diplomatic, government entities with evolving toolset*. Milpitas, CA: Mandiant Intelligence. Available at: <https://www.mandiant.com/resources/blog/apt29-continues-targeting-microsoft> (Accessed: 15 September 2025).

ALPHV/BlackCat Ransomware FBI (2023) *ALPHV BlackCat ransomware indicators of compromise*.

Federal Bureau of Investigation Flash Report CU-000175-MW. Washington, DC: FBI Cyber Division.

Regulatory and Compliance

GDPR and Identity Management European Union Agency for Cybersecurity (2023) *Guidelines on identity and access management for cloud computing*. Heraklion: ENISA. Available at:

<https://www.enisa.europa.eu/publications/security-aspects-of-cloud-computing> (Accessed: 29 May 2025).

Healthcare Compliance U.S. Department of Health and Human Services (2024) *HIPAA security rule guidance*. Washington, DC: HHS Office for Civil Rights. Available at: <https://www.hhs.gov/hipaa/for-professionals/security/index.html> (Accessed: 14 July 2025).